

A Low-Cost Glasses-Based Brain-Computer Interface Prototype for Real-Time Emotion and Intent Recognition

Multimodal Wearable Neural Technology Utilizing Electroencephalography (EEG), Electrooculography (EOG), and Electromyography (EMG) Signals

Abstract

This study proposes a glasses-based BCI prototype system that captures frontotemporal biosignals through six dry electrodes, with signals containing EEG, EOG, and EMG components. Through a single-participant pilot study, the system evaluated two core tasks: emotion recognition triggered by auditory stimuli and intent recognition through gaze direction and jaw clenching. Task-specific neural network models deployed on Raspberry Pi 5 demonstrated the feasibility of real-time processing. The system achieved 73.15% accuracy in four-class intent recognition tasks and 67.90% accuracy in three-class emotion recognition tasks, with inference times of 3.40 ms and 0.16 ms respectively. By integrating the practicality of dry electrodes, the efficiency of embedded hardware, and custom deep learning models, this study provides a practical solution for wearable BCI technology, enabling volitional control and emotional perception interaction in everyday environments.

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1. Introduction

1.1. Research Motivation and Background

Human emotions profoundly influence communication, memory, and decision-making. Traditional human-computer interfaces mainly respond to explicit commands, often ignoring users' emotional states, resulting in rigid interactions. Affective computing aims to enable machines to perceive and respond to emotional cues, thereby improving interactive experiences [1]. Meanwhile, for patients with neuromuscular diseases, more intuitive and diverse control methods are crucial for device operation and environmental interaction.

Brain-computer interfaces (BCIs) convert neural activity into control commands for computers and assistive devices, providing a viable path for the above problems [2]. Electroencephalography (EEG) can record brain activity through scalp electrodes with millisecond-level temporal resolution, capturing neural oscillations related to cognition and emotion; commonly used frequency bands include δ (0.5-4 Hz), θ (4-8 Hz), α (8-13 Hz), β (13-30 Hz), and γ (>30 Hz) [3]. Research also shows that stronger left frontal activation (lower α power) is commonly associated with positive, approach-oriented emotions, while stronger right frontal activation corresponds to negative, avoidance-oriented emotions [4].

Early BCIs relied on wet electrodes and laboratory-grade equipment, requiring skin preparation and conductive gel, with gel drying over time leading to increased impedance and limited recording duration, plus inconvenient cleaning procedures [5]. Although consumer-grade EEG headsets and open-source hardware have promoted research democratization, existing devices mostly focus on single tasks like meditation or gaming, rarely considering multimodal integrated applications of "emotion recognition + intent recognition"; and commercial systems remain expensive and bulky.

To address these limitations, advances in dry electrodes bring new opportunities for wearable BCIs: no gel required, convenient wearing, but need to address higher contact impedance and greater sensitivity to motion artifacts [6]. More importantly, signals traditionally regarded as "artifacts" in EEG analysis can in fact provide effective control information. For instance, horizontal eye movements generate strong, low-frequency EOG components that dominate the frontal EEG channels [7], while jaw clenching produces high-amplitude EMG activity extending into the 20-200 Hz range [8]. These signals can be consciously utilized as control channels. Additionally, open hardware like PiEEG-16 combines Raspberry Pi with multi-channel

EEG/EMG/ECG acquisition and on-device feature/classification, validating the feasibility of efficient neural signal classification on embedded processors [9].

This project targets the integrated goal of "intent recognition + emotion recognition", utilizing EEG, EOG, and EMG signals for parallel modelling. The design incorporates six dry electrodes integrated into glasses frames, covering F7/FT7/T3 and F8/FT8/T4, capturing lateral frontal and temporal activity related to valence, gaze direction, and jaw clenching. This design leverages the natural form of glasses to provide a comfortable, minimally invasive interface, avoiding the bulkiness of traditional caps. In addition, the system achieves real-time inference on embedded hardware, supporting emotion-aware context sensing and user intent recognition, targeting applications such as emotion-adaptive smart home control and assistive communication. The system overview is shown in *Fig 1*.

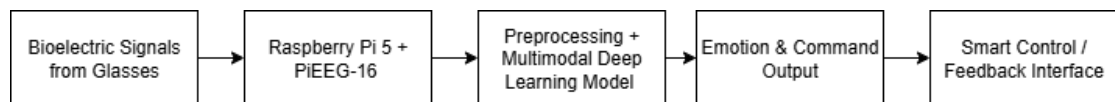


Fig 1. System Overview and On-Device Inference Pipeline (this project)

1.2. Problem Statement and Challenges

The development of a glasses-based BCI prototype entails addressing three primary technical challenges:

1. **Balancing wearability and signal quality.** Wet electrodes require conductive gel and lengthy skin preparation procedures, while dry electrodes, though more convenient, exhibit higher contact impedance and are susceptible to motion artifacts. Even advanced spring-loaded designs face difficulties maintaining stable recording under gel-free conditions [5]. Achieving reliable acquisition with as few dry electrodes as possible in a glasses frame requires precise mechanical structures and effective electromagnetic shielding.
2. **Separation and task-adaptive processing of multimodal signals.** Lateral temporal electrodes simultaneously capture EEG, EOG, and EMG signals. EOG is predominantly low-frequency, while EMG activity is mainly concentrated in higher frequency bands [7]. These signals may mask subtle neural oscillations needed for emotion recognition but are valuable control information in intent recognition. Therefore, preprocessing pipelines must be task-adaptive: suppressing eye movement and muscle artifacts when inferring emotional valence, while preserving relevant signals when recognizing gaze

or jaw movements.

3. **Real-time embedded processing capability.** Executing multimodal fusion and classification on embedded platforms faces strict latency and power consumption constraints. While low-cost BCI systems have achieved high emotion classification accuracy through neural networks [9], how to maintain real-time performance while fusing multimodal information remains insufficiently explored. This requires designing efficient network architectures and utilizing optimization methods like model compression to meet hardware constraints.

1.3. Research Objectives and Contributions

To address the above challenges, this study sets four core objectives for the glasses-based BCI prototype:

1. Integrate a six-electrode dry sensor array into eyeglass frames for comfortable, stable acquisition.
2. Apply a unified preprocessing pipeline (50 Hz notch, 1–40 Hz bandpass, z-score normalization) for both tasks.
3. Build multi-branch neural networks (temporal/frequency/time–frequency) and compress them for embedded deployment.
4. Release code, trained models, and results; withhold raw biosignals for privacy.

By demonstrating real-time emotion and intent recognition capabilities on low-power hardware, this study aims to advance wearable BCI technology and provide support for emotional assistance and interaction technologies.

1.4. Structure of the Dissertation

The remainder of this study is organized as follows:

- Chapter 2 reviews related work on EEG-based BCIs, emotion recognition, and intent recognition, emphasizing current progress and existing limitations.
- Chapter 3 presents research methods, including hardware design, experimental tasks, and algorithm implementation.
- Chapter 4 reports system implementation and experimental result analysis.
- Chapter 5 discusses research findings, system limitations, potential applications, and future directions.

- Chapter 6 concludes the dissertation.

2. Literature Review

2.1. EEG-based Brain-Computer Interface System

2.1.1. Evolution of BCI Systems

Modern BCI research can be traced back to early explorations of event-related potentials in the 1970s-1980s. Farwell and Donchin [10] introduced the P300 “speller,” showing that users could select characters by attending to rare visual stimuli in a letter matrix. Wolpaw et al. [2] summarized the first decade of systematic BCI studies, reviewing slow cortical potentials, sensorimotor rhythms, and event-related potentials, and highlighting their potential in restoring communication for patients with severe motor disabilities. Later, Serruya et al. [11] provided one of the first demonstrations that cortical activity could be used in real-time to control an external device, marking the rise of invasive BCI research.

In contrast, non-invasive systems are more suitable for widespread applications. In the 2000s, commercial headsets such as Emotiv and NeuroSky appeared. Nijholt et al. [12] reviewed these early consumer-grade systems, noting their limited number of channels and reliance on proprietary software. More recently, open-source hardware and single-board computers have expanded BCI applications into everyday scenarios. Rakhmatulin et al. [9] introduced *PiEEG-16*, a low-cost Raspberry Pi shield capable of acquiring up to 16 biosignals (EEG, EMG, ECG) and performing real-time processing on the device. The design is open-source and priced at about \$390, excluding the Raspberry Pi board. Alongside *PiEEG-16*, community platforms such as *JNEEG* integrate analog front-ends with Raspberry Pi or Jetson Nano, supporting real-time EEG/EMG/ECG acquisition and classification [13]. These innovations democratize BCI research, though most implementations still emphasize single-task emotion recognition, with limited attention to intent recognition or alternative wearable form factors.

2.1.2. Wearable EEG Technologies and Electrodes

Sensor technology is central to wearable BCIs. Traditional wet Ag/AgCl electrodes provide low impedance and high signal fidelity but require conductive gel and scalp preparation, with gel drying limiting recording duration. Dry electrodes offer greater convenience by eliminating gel, and multipin designs have shown stable acquisition validated against wet electrodes [14]. However, they typically exhibit higher impedance and stronger motion artifact sensitivity. Comprehensive reviews [8]

summarize developments such as silicon-based dry sensors and wireless wearable EEG systems, emphasizing that while dry electrodes promise to extend EEG beyond laboratories, advanced signal processing remains essential to mitigate their inherent shortcomings.

2.1.3. Limitations of Current Commercial Solutions

Existing commercial BCI products mainly target gaming, meditation, or basic control tasks, with user access to raw data typically restricted. They often provide only a few electrodes and do not follow the international 10-20 system [15] layout (as shown in *Fig 2*), limiting spatial resolution and interpretability. Some devices rely on cloud processing, increasing latency and raising privacy concerns.

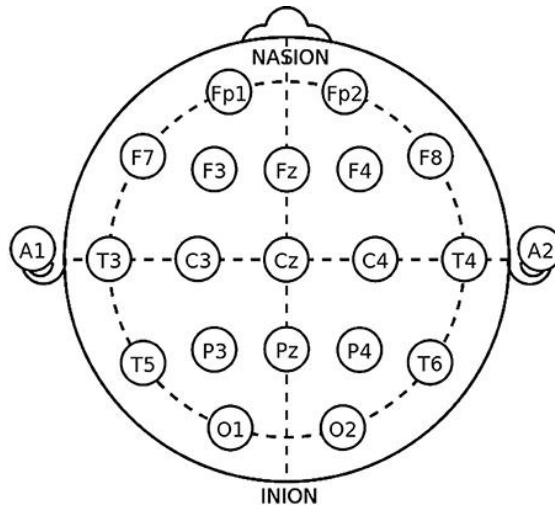


Fig 2. International 10-20 System ($T3 \equiv T7$, $T4 \equiv T8$); adapted from Rojas et al. [16].

Mohammed Mosa and Harris [17] used Emotiv headsets with Raspberry Pi to achieve home automation, allowing disabled users to control lighting and fans. Their study shows these systems have potential but also limitations including high cost, proprietary nature, single functionality, and lack of emotion recognition.

2.2. Emotion Recognition Based on Physiological Signals

2.2.1. Multimodal Emotion Recognition

Emotion recognition relies on fusion of neural, muscular, autonomic nervous, and behavioural signals. The DEAP dataset proposed by Koelstra et al. [18] recorded EEG (see *Fig 3*), skin conductance, respiration, and EMG synchronously from 32 participants watching 40 music videos, with subjective ratings of segments. Analysis showed that combining EEG with peripheral signals can significantly improve classification performance.

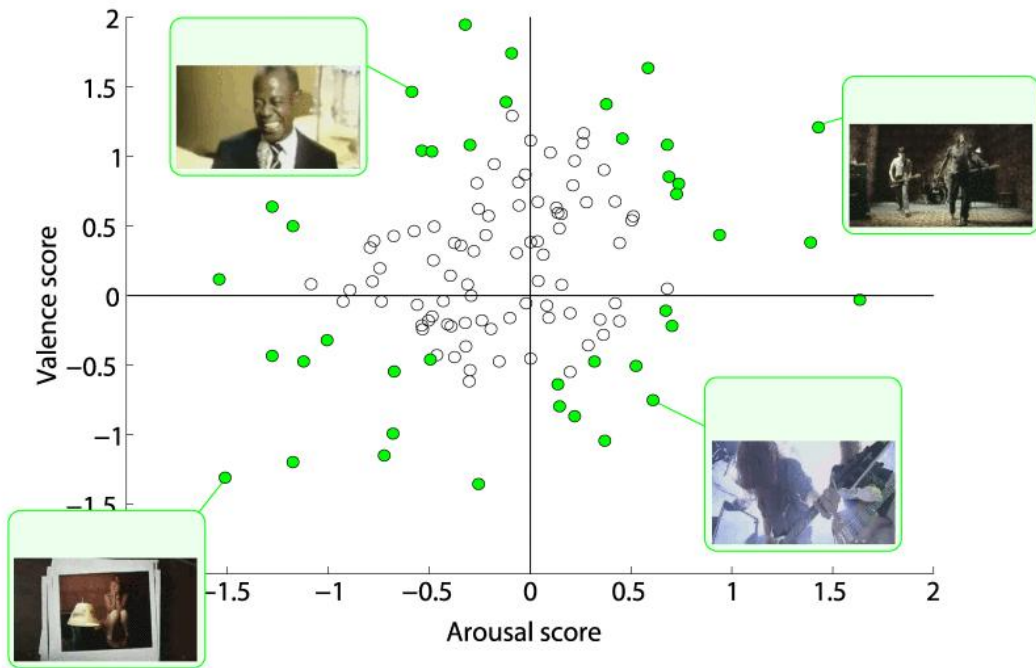


Fig 3. Distribution of video stimuli in the valence-arousal space, with sample frames shown, adapted from Koelstra et al. [18].

Beyond EEG-peripheral combinations such as DEAP, researchers have also explored integrating EEG with fNIRS and facial video to enhance emotion recognition. The EEGLAB toolbox [19] has become a cornerstone for EEG preprocessing, particularly for independent component analysis (ICA). Nevertheless, EEG–video fusion faces key challenges: facial expressions can be consciously regulated, and strong inter-subject variability further limits model generalization [7] [8].

2.2.2. Neurophysiological Basics of Emotion Decoding

Emotion processing involves widespread brain regions, with frontal α asymmetry playing a particularly central role. Davidson [20] summarized the link between enhanced left frontal activation (reflected in reduced α power) and positive, approach-related emotions, and between enhanced right frontal activation and negative, avoidance-related emotions. Typical EEG analyses compute power spectral density in relevant frequency bands and extract time–frequency features. Event-related potentials are also commonly identified. ICA, popularized by the EEGLAB toolbox [19], advanced the separation of neural sources from artifacts, yet distinguishing oscillatory brain activity from EOG or EMG contamination still demands fine preprocessing. Rashid et al. [8] further noted that advanced approaches like blind source separation and canonical correlation analysis often incur high

computational overhead, restricting their use in real-time BCI systems.

2.2.3. Available Datasets and Limitations

Multiple datasets support EEG-based emotion recognition, including MAHNOB-HCI, SEED, AMIGOS, and DREAMER, which typically combine EEG with peripheral signals such as ECG, skin conductance, and respiration, along with self-assessment labels. Katsigiannis and Ramzan [21] introduced the DREAMER dataset, recording EEG and ECG from wireless low-cost devices to support affective computing research.

However, most datasets were acquired in controlled laboratories using wet electrodes and do not contain explicit control actions. The lack of wearable scenario corpora that simultaneously record emotions and intentions limits evaluation of related algorithms [8].

2.3. Intent Recognition and Control Interfaces

2.3.1. Physiological Control Signals

Intent recognition commonly uses paradigms like motor imagery, P300, or SSVEP. Eye and muscle signals provide more intuitive alternative control pathways. Saha and Baumert [7] reviewed methods combining eye tracking with EEG, noting that horizontal eye movements produce EOG <30 Hz. Rashid et al. [8] reported that jaw activity produces high-amplitude EMG at 20-200 Hz, much higher than EEG, requiring task-adaptive filtering.

These signals can serve as command inputs while also contaminating emotion recognition frontal/temporal channels. Jiang et al. [22] recommended hybrid strategies combining regression, filtering, and blind source separation to suppress artifacts. Especially when the same signal is both noise and control information, whether to remove it becomes crucial.

2.3.2. Assistive Technology and Smart Home Applications

BCIs have been applied to wheelchair navigation, robotic arm control, and smart homes. The low-cost home automation system by Mohammed Mosa and Harris [17] allows disabled users to control lighting and fans through Emotiv and Raspberry Pi, demonstrating the technology's potential. The authors estimate that at least one billion people globally could benefit from such solutions.

However, they also noted that current devices are expensive, rely on proprietary software, and lack emotion recognition. If emotion recognition could be integrated, systems could adjust lighting or music based on emotions and monitor frustration and fatigue to improve safety. How to simultaneously achieve intent recognition and

emotion recognition in a single wearable device remains an open question.

2.4. Research Gaps and Motivation

The literature review reveals the following research gaps:

1. Lack of multimodal wearable BCIs that integrate EEG, EOG, and EMG into comfortable forms like glasses; most systems only support emotion recognition or control functions, rarely both.
2. Existing datasets rarely include both emotion and intent annotations simultaneously, limiting algorithm training and evaluation [18].
3. Most signal processing pipelines treat eye movements and muscle electrical activity as artifacts rather than potential control cues; adaptive preprocessing strategies based on task context are lacking [22].
4. Real-time execution of multimodal fusion on embedded hardware remains challenging. Although low-cost systems have achieved high emotion classification accuracy [9], real-time multimodal fusion under power and latency constraints has not been widely validated.

Addressing these issues is key to advancing BCIs from laboratory to practical, multifunctional wearable systems.

3. Methodology

3.1. System Overview

Brain-computer interface (BCI) systems have traditionally been developed in controlled laboratory environments, often using wet electrodes and desktop amplifiers. This project targets everyday applications, proposing a glasses-based BCI: six dry electrodes embedded in the frame (F7/FT7/T3, F8/FT8/T4) with on-device inference completed on Raspberry Pi 5.

Bioelectric signals are acquired by PiEEG-16 HAT (ADS1299 dual-chip). Although PiEEG-16 supports 16 channels, the prototype only enables 6 channels near the temples to capture frontotemporal activity related to emotional valence, gaze actions, and jaw EMG. The system targets two types of tasks: intent recognition (baseline, jaw clench, left gaze, right gaze) and emotion recognition (negative, neutral, positive), using late fusion multimodal networks that fuse temporal,

frequency, and time-frequency representations.

To approach real deployment constraints, the pipeline emphasizes reproducible on-device processing rather than pursuing extreme metrics. Sampling rate is 250 Hz; preprocessing consists sequentially of 50 Hz notch, 1-40 Hz bandpass, and z-score normalization. Three branches in parallel:

- Temporal branch (1D-CNN + LSTM) processes normalized microvolt sequences.
- Frequency branch calculates log-scaled relative band power ($\theta/\alpha/\beta$) from Welch periodograms, normalized by 1-40 Hz total power.
- Image branch generates spectrograms (single-channel ResNet-18).

Finally, late fusion classifiers integrate branch embeddings.

The same architecture serves both tasks, but final models are pruned by task: intent recognition uses temporal and frequency, emotion recognition uses only frequency features. Real-time parameters: 2.0 s windows, 1.0 s update intervals (50% overlap).

As shown in *Fig 4*, PiEEG-16 (ADS1299) provides 6-channel EEG, on-device preprocessing (50 Hz notch, 1-40 Hz bandpass, z-score), and through parallel temporal/frequency/image branches with late fusion achieves real-time 4-class intent and 3-class emotion prediction.

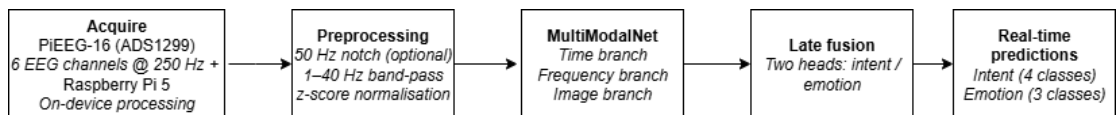


Fig 4. On-device Inference Pipeline for Glasses-based BCI

3.2. Hardware Design

3.2.1. Dry Electrode Glasses Prototype

The physical carrier is a 3D-printed frame with 6 dry electrodes installed on the inner rim. Electrodes are placed according to the 10-20 system [15] at left F7, FT7, T3 and right F8, FT8, T4 (T3/T4 $\equiv$ T7/T8; see *Fig 5*).

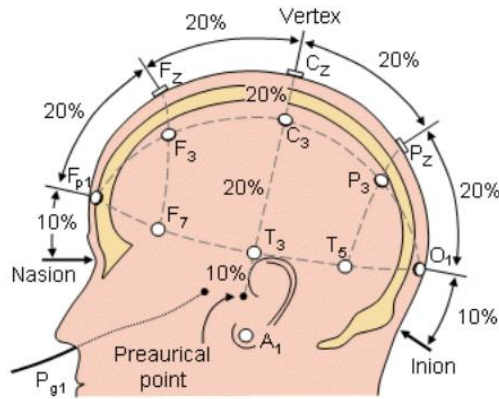


Fig 5. 10-20 map highlighting F7/FT7/T3 and F8/FT8/T4 ($T3 \equiv T7$, $T4 \equiv T8$ in legacy labels), adapted from Orban et al. [23].

The frontotemporal positions were selected to capture cortical rhythms related to emotional valence [4] and intentionally record eye movement and muscle components useful for intent recognition. The frame is customized to the participant to ensure stable dry electrode-skin contact, reducing impedance fluctuations and motion artifacts (*Fig 6* shows the participant wearing the prototype, *Fig 7* shows the SolidWorks CAD model).



Fig 6. Participant wearing the prototype; dry electrodes in contact with the skin.



Fig 7. SolidWorks model of the 3D-printed glasses showing eight electrode mounts (six mounts are used).

3.2.2. PiEEG-16 and Raspberry Pi 5

Six electrodes connect to PiEEG-16 (official Raspberry Pi HAT form factor PCB) with two Texas Instruments ADS1299 analog front-ends onboard (see Fig 8). Each ADS1299 provides 8-channel, programmable gain, synchronous 24-bit Δ - Σ conversion. The prototype enables 6 channels (F7/FT7/T3, F8/FT8/T4).

ADS1299 is a widely used analog front-end in research-grade BCIs thanks to its low-noise, highly integrated design — offering 8 simultaneous sampling channels and 24-bit resolution, with a built-in programmable gain amplifier and on-chip reference and oscillator. Experimental validation by Rashid et al. [24] found that ADS1299-based systems using wet Ag/AgCl electrodes perform comparably to laboratory-grade amplifiers in low-frequency EEG (< 40 Hz)

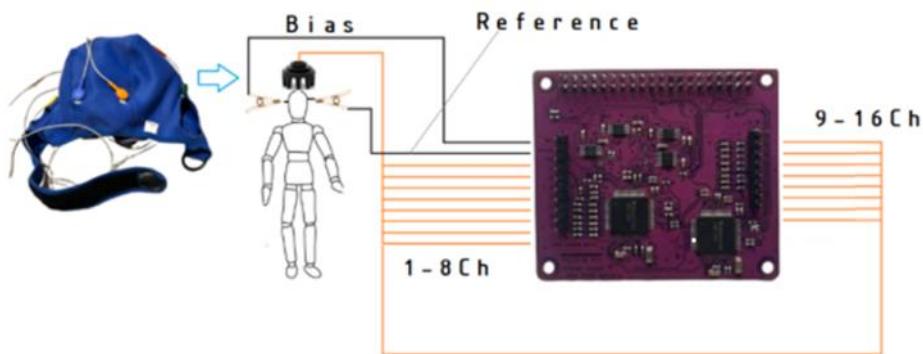


Fig 8. PiEEG-16 general schematic, adapted from Rakhmatulin, I. [9].

Acquisition is implemented by the project's ADS1299Controller: at startup, Raspberry Pi 5 configures PiEEG-16 via SPI, putting ADS1299 into continuous

conversion mode at 250 Hz sampling rate with target channels enabled. Each cycle parses 24-bit two's complement frames and converts to microvolts according to full-scale factors; basic outlier checking ensures frame integrity (assembly structure shown in *Fig 9*).

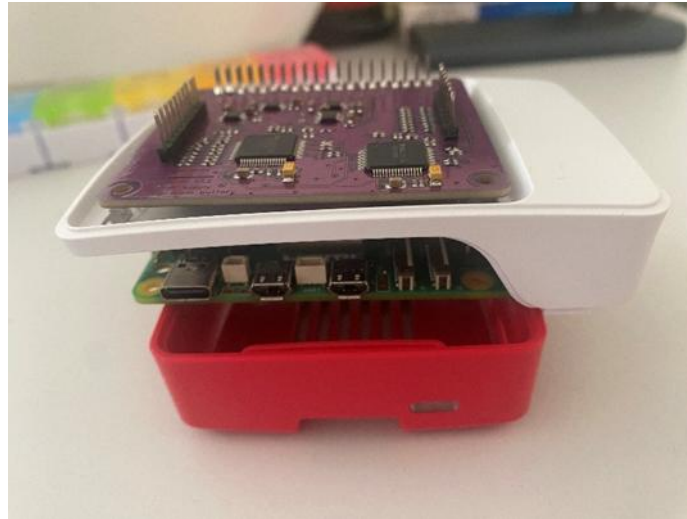


Fig 9. Assembled Raspberry Pi 5 with PiEEG-16 HAT used for signal acquisition.

3.2.3. Complete Prototype Assembly

The prototype integrates dry electrode glasses, PiEEG-16, and Raspberry Pi 5 into a compact, low-cost, minimal-cable system (see *Fig 10*). To reduce conducted and coupled noise, the display is powered by independent battery, Raspberry Pi 5 and PiEEG-16 use portable power, peripherals use wired mouse/keyboard to reduce wireless interference, and hardware Bluetooth/Wi-Fi are disabled. Recording is conducted in low-noise environments. This assembled system demonstrated stable and reliable operation, serving as the hardware foundation for the acquisition protocol below; Chapter 4 will combine preprocessing results to provide short-circuit baseline noise characterization for quantifying input reference noise.

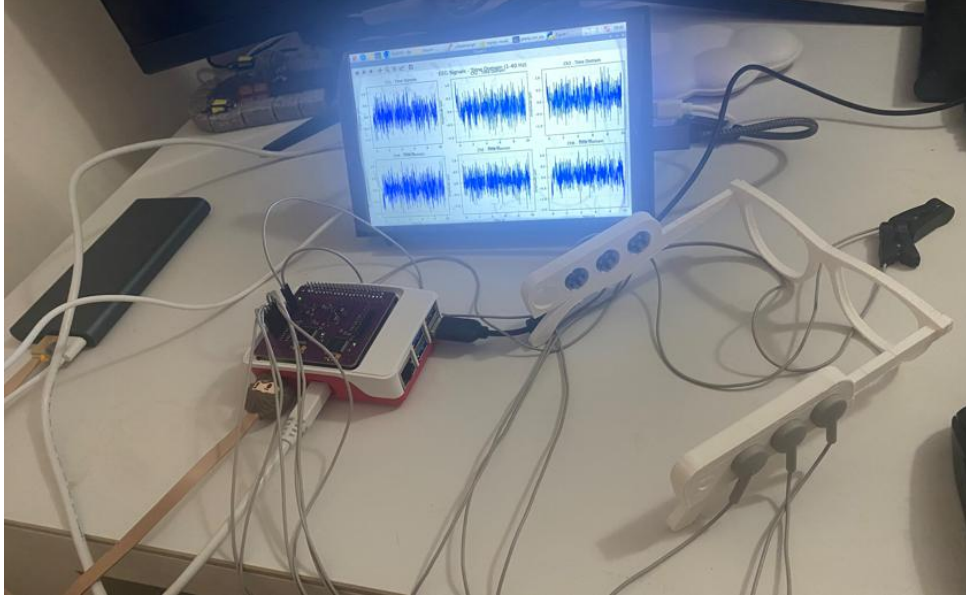


Fig 10. Complete system setup (glasses, PiEEG-16, Raspberry Pi 5, independent display power, wired peripherals).

3.3. Data Acquisition

3.3.1. Intent Acquisition Task

In this study, each intent trial followed a three-phase protocol: Baseline (3 s), Action (5 s), and Recovery (2 s), resulting in a total duration of approximately 10 s (2,500 samples at 250 Hz). This setup was chosen to match the downstream windowing strategy. The recorded data from each trial were stored in a single CSV file containing the following columns: timestamp, phase_time, phase, and six channels (ch1–ch6).

Session-level metadata (sampling rate, channel names, timing markers, participant) are stored separately. Filenames follow `intent_trial_<participant>_<trialIndex>_<action>_<timestamp>.csv`, and are parsed downstream to obtain the label set {baseline, jaw_clench, gaze_left, gaze_right}.

The CSV includes per-sample phase and phase_time, enabling phase-aware selection; each trial was segmented into 2.0-second windows with a 1.0-second hop for model training, using window-level supervision.

3.3.2. Emotion Acquisition Task

The emotion task was conducted following the DEAP protocol [18], in which 30-second auditory stimuli were used to induce negative, neutral, and positive valence. To achieve a balance between sufficient emotional induction and real-time

applicability, only the final 10 seconds of each stimulus (i.e., starting at 20 s) were recorded, sampled at 250 Hz. During the task, the interface displayed only a timer while the audio was played externally. The recorded data from each trial were stored in a single CSV file containing the following columns: timestamp, recording_time, and six channels (ch1–ch6).

File headers remain concise (no inline JSON). Post-trial optional acquisition of 1-9 SAM self-assessment rating [25]. In this project, scores are discretized to {negative, neutral, positive} and both discrete categories and numerical scores are recorded in filenames.

3.3.3. Dataset Construction

All CSV files and metadata were parsed to construct model-ready datasets. Labels were derived from filenames, corresponding to {baseline, jaw_clench, gaze_left, gaze_right} for intent recognition and {negative, neutral, positive} for emotion recognition. The channel order was fixed as [F7, FT7, T3, F8, FT8, T4], with a sampling rate of 250 Hz.

Each trial is segmented into 2.0-second windows with 1.0-second step (500/250 sample points), evaluated at window level. The final dataset comprised 648/144/108 windows for intent recognition (train/validation/test) and 378/81/81 windows for emotion recognition.

The resulting datasets comprise time-domain arrays (channels $\times$ samples) together with class labels, channel names, sampling rate, and task annotations specifying whether the trial corresponds to intent detection or emotion recognition.

The dataset was first split into training and temporary sets using a stratified 70/30 split on trials. The temporary set was then evenly divided (50/50) into validation and test sets. This procedure resulted in final distributions of 72/16/12 trials for intent recognition and 42/9/9 trials for emotion recognition.

In addition to the fixed train/validation/test split, five-fold cross-validation [26] was later applied to further assess model robustness.

3.4. Signal Processing and Feature Extraction

3.4.1. Preprocessing

EEG signals are susceptible to noise from power line interference, slow drift, and muscle artifacts, making preprocessing a necessary first step [27]. A flexible preprocessing pipeline was implemented, comprising the following steps:

1. **Notch filtering:** Apply second-order IIR notch filter at 50 Hz using

`scipy.signal.iirnotch` with quality factor $Q=25$ (adjustable to 60 Hz for regions). This narrow-band filter attenuates power line interference while preserving adjacent frequency content. The filter is applied using zero-phase filtering (`filtfilt`) to avoid phase distortion.

2. **Bandpass filtering:** Each channel passes through a fourth-order Butterworth filter implemented as second-order sections (SOS) for numerical stability. The 1-40 Hz passband removes DC drift and slow motion artifacts below 1 Hz while rejecting high-frequency noise and aliasing above 40 Hz. The SOS implementation of a fourth-order Butterworth filter (`scipy.signal.butter`, `scipy.signal.sosfiltfilt`) ensures greater numerical stability compared to traditional transfer function methods.
3. **Z-score normalization:** Within each window, channels are normalized: $z = (x - \mu)/\sigma$, where μ is channel mean, σ is standard deviation, with small ϵ ($1e-8$) added to prevent division by zero. This normalization ensures consistent amplitude scales across recording sessions and participants.

3.4.2. Time Domain Features

The first branch of the multimodal network processes raw time-domain sequences. This branch contains two one-dimensional convolutional layers (1D-CNN) that extract short-term patterns over 500-sample windows. Each convolutional layer is followed by ReLU activation and max pooling [28], with resulting feature maps subsequently input to a Long Short-Term Memory network (LSTM) to capture temporal dependencies [29].

- In intent recognition, LSTM can distinguish sustained jaw clench signals from transient eye movements.
- In emotion recognition, LSTM can capture slow rhythmic oscillations related to emotional valence.

Since this branch directly processes Z-score standardized microvolt sequences, no hand-crafted features are required.

3.4.3. Frequency Domain Features

The second branch summarizes frequency-domain information for each 2-second window by computing relative band power. The `bandpower` function in `features.py` uses Welch's method [30] to calculate power spectral density and integrates over classical EEG bands: θ (4-8 Hz), α (8-13 Hz), β (13-30 Hz). Each band power is normalized by 1-40 Hz total power, resulting in a matrix of six channels $\times$ three bands, reflecting relative energy distribution across neural rhythms.

- In emotion recognition, frontal α asymmetry (i.e., F7 vs F8 α power differences) is typically related to approach/avoidance tendencies [31]. While this pilot study does not explicitly compute this metric, the network can learn similar patterns from relative power.
- In intent recognition, jaw clenching generates strong β wave energy in temporal electrodes.

This branch uses two convolutional layers and global average pooling to convert frequency-domain features into fixed-length vectors.

3.4.4. Spectrogram Images

The third branch uses time-frequency spectrograms as input to convolutional neural networks. The `spectrogram_images` function divides each 2-second window into overlapping segments, applies Short-Time Fourier Transform (STFT) [32], and computes log power spectra [33]. Generated spectrograms are Z-score standardized and resized to fixed input dimensions.

These images are processed by modified ResNet-18 [34]: the first convolutional layer is changed to single-channel input, and the final fully connected layer is replaced with low-dimensional embedding. Spectrograms can reveal non-stationary patterns, such as β activity bursts during jaw clenching, or α and β band transitions in different emotional states.

Because STFT and image processing incur substantial computational overhead, this branch can be disabled to trade performance for latency. In configurations where channels are averaged prior to STFT execution, a single spectrogram is computed for each window.

3.4.5. Data Integrity and Signal Quality Assessment

Signal quality is checked before feature extraction to ensure classification reliability. For each 2-second window, after bandpass (1-40 Hz) and 50 Hz notch filtering but before Z-score normalization, root-mean-square (RMS) amplitude and peak-to-peak range are calculated per channel, with amplitudes reported in microvolts (μ V).

Additionally, signal quality is further assessed through power spectral checks and monitoring for typical artifacts, including drift, saturation, broadband noise. PiEEG-16 with ADS1299 provides 24-bit Δ - Σ conversion suitable for microvolt-level EEG acquisition (Texas Instruments, 2015).

3.4.6. Feature Space Analysis Methods

To evaluate feature space structure and class separability, this study employs

two dimensionality reduction methods:

Principal Component Analysis (PCA): Used for linear dimensionality reduction [35]. Time-domain features (500 points $\times$ 6 channels) are concatenated with band power features (3 bands $\times$ 6 channels) to form high-dimensional vectors. PCA extracts orthogonal components that maximize variance through covariance matrix eigen decomposition, with each component's variance contribution reflecting feature space dimensional structure.

t-SNE: Used for nonlinear dimensionality reduction visualization [36]. This method minimizes KL divergence between similarity distributions in high-dimensional and low-dimensional spaces:

$$KL(P \parallel Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

Where p_{ij} represents similarities in the original space and q_{ij} in the embedded space using Student's t-distribution. t-SNE with perplexity 30 and 1000 iterations is applied to visualize whether classes form distinct clusters in the feature space, providing insight into the separability achievable by non-linear classifiers.

3.5. Multimodal Network Architecture and Training

3.5.1. Network Architecture

The MultiModalNet classifier integrates three parallel feature streams. As shown in Fig 11, each branch outputs 64-dimensional latent embeddings.

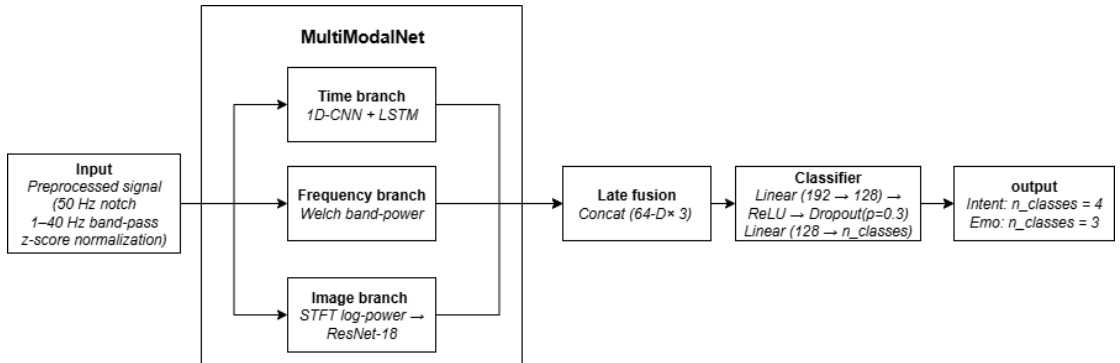


Fig 11. MultiModalNet architecture with late-fusion classifier

Temporal branch: Consists of two layers of 1D convolution and max pooling, followed by LSTM, with final hidden state as temporal embedding.

Frequency branch: Processes band power through pointwise convolution and global average pooling to obtain compact spectral representation.

Image branch: Based on single-channel ResNet-18, accepts spectrogram input, with final fully connected layer changed to output 64-dimensional embedding.

Three branch embeddings are concatenated into 192-dimensional vectors, then reduced to 128 dimensions through fully connected layers, followed by ReLU activation [28], 0.3 probability dropout, and finally mapped to task class numbers (intent 4 classes, emotion 3 classes).

3.5.2. Loss Functions and Optimization

Training uses class-weighted cross-entropy with label smoothing ($\epsilon = 0.1$) to handle class imbalance and reduce overfitting [37].

A two-stage training strategy is adopted:

Architecture comparison stage: Uses Adam optimizer [38] for rapid iteration on candidate models.

Complete training stage: Trains under AdamW optimizer [39] with ReduceLROnPlateau learning rate scheduling, combined with weight decay to enhance generalization capability.

The training objective was a cross-entropy loss [37] extended with class weights for imbalanced learning [40] and label smoothing [37]:

$$L = \frac{-1}{N} \sum_{i=1}^N w_{y_i} \sum_{c=1}^C q_c^{(i)} \log p_c^{(i)}, \quad q_c^{(i)} = \begin{cases} 1 - \epsilon, & c = y_i \\ \frac{\epsilon}{C - 1}, & c \neq y_i \end{cases}$$

Where $p_c^{(i)}$ is the softmax probability for class c of sample i ; w_{y_i} is the class weight of the ground-truth class [40].

3.5.3. Performance Metrics

Evaluation follows standard metrics for BCI systems, emphasizing transparency rather than inflating accuracy:

Accuracy: The proportion of correctly classified samples [41]:

$$Accuracy = \frac{n_{correct}}{n_{total}}$$

Macro-averaged F_1 Score: The harmonic mean of precision and recall, averaged across all classes without weighting [42]:

$$F1 - macro = \left(\frac{1}{C}\right) * \sum_{c=1}^C \frac{2 \cdot Precision_c \cdot Recall_c}{Precision_c + Recall_c}$$

where C is the number of classes, and for each class c :

$$Precision_c = \frac{TP_c}{TP_c + FP_c}, \quad Recall_c = \frac{TP_c}{TP_c + FN_c}$$

Cohen's Kappa: A statistic that measures inter-rater agreement, accounting for chance agreement [43] [44]:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o is the observed agreement (accuracy) and p_e is the expected agreement by chance:

$$p_e = \sum_{c=1}^C P_{c,pred} \cdot P_{c,true}$$

with $P_{c,pred}$ and $P_{c,true}$ being the marginal probabilities for class c in predictions and true labels.

Information Transfer Rate (ITR): A measure of the communication bandwidth of the BCI system [45]:

$$ITR = \log_2(N) + P \log_2 P + (1 - P) \log_2 \left(\frac{1 - P}{N - 1} \right)$$

$$ITR(bits/min) = ITR(bits/trial) \times \frac{60}{T}$$

where N is the number of classes, P is the classification accuracy, and T is set by the inference update interval ($T = \text{hop_size} / \text{sampling_rate}$), consistent with the sliding-window pipeline. In this study, $T = 1.0$ s for ITR reporting (hop = 250 samples at 250 Hz); the real-time also used a 1-second update interval.

Additionally, confusion matrices are plotted to observe inter-class confusion [46], and inference time, throughput, and real-time factor are reported.

3.5.4. Ablation Experiments

Model selection uses a two-stage procedure:

Stage 1 (rapid comparison): Train four candidate architectures under identical settings—temporal-only, frequency-only, image-only, and complete multimodal network (temporal + frequency + image). Each candidate uses Adam (learning rate 1×10^{-3}) optimized for 10 epochs on the same train/validation split. Validation accuracy serves as the primary selection criterion; Macro F1, Cohen's Kappa, balanced accuracy, and ITR are computed and recorded for completeness.

Stage 2 (complete training): Selected architecture is reinitialized and trained using AdamW [39] (learning rate 3×10^{-4} , weight decay 1×10^{-4}) with ReduceLROnPlateau scheduling for 50 epochs. Class-weighted cross-entropy with label smoothing ($\varepsilon = 0.1$) is always used. Output heads are sized by task (intent 4 classes; emotion 3 classes). Section 4.3 reports quantitative differences among the four candidates and computational profiles of selected models.

3.6. Real-Time Implementation and Integration

The ultimate goal is to provide continuous feedback on users' cognitive and emotional states. A dedicated real-time prediction module was deployed on Raspberry Pi, integrating acquisition, preprocessing, and classification. It instantiates SignalAcquisition objects, preloads trained models, and sets up deque buffers to store the latest 2 seconds of data per channel.

Every 1 second, the predictor copies the current buffer, applies bandpass filtering and z-score normalization, computes band power (optional spectrogram), and runs inference. Predicted intent and emotion classes map to simple commands: baseline or neutral trigger no actions; jaw clench may toggle switches; left/right gaze can scroll menus; positive/negative emotions may adjust brightness or music volume. Majority voting of the most recent three windows smooth predictions to reduce jitter.

Using Python's multiprocessing module, data acquisition and inference run on independent threads to prevent blocking. Preliminary measurements indicate end-to-end latency below 200 ms per window, well below perceptual thresholds for real-time feedback [47]. Device drivers are not enabled in this prototype; simple command mapping is reserved for future integration. Detailed results are in Section 4.5.

The complete implementation (code, results) is available at the project repository; the URL is provided in the **Appendix**.

3.7. Ethics and Data Governance

Physiological signals used in this project were all collected by the author as part of a self-experimentation protocol. No external participants were involved, and no personally identifiable information was collected. Collected data is privately held by the author and not publicly released due to its biometric attributes. However, complete code, trained models, and experimental results are open-sourced to ensure transparency and reproducibility. The project follows best practices for responsible data handling and scientific integrity in BCI research [48].

4. Experimental Results and Analysis

This chapter presents key data and results from the study. Detailed implementation outputs, including preprocessing logs, class distributions, training progress, and evaluation metrics, are provided in the **Appendix**.

4.1. Overview of Data Acquisition and Dataset Size

This project used six dry electrodes (F7/FT7/T3 and F8/FT8/T4), sampled at 250 Hz, using ADS1299-based analog front-end. Each trial lasted approximately 10 seconds (2,500 samples per recording). Acquisition included two tasks: intent recognition (baseline, jaw clench, left gaze, right gaze) and emotion recognition (negative, neutral, positive).

Short-circuit noise hardware self-test was completed before formal acquisition. Monitoring-only 1-40 Hz bandpass filtering was used on the plotting end; this visual filter is not part of experimental preprocessing. Traces are in μV range with means less than 1 μV , showing good consistency across channels, indicating stable analog front-end and grounding system meeting requirements for subsequent acquisition (see Fig 12).

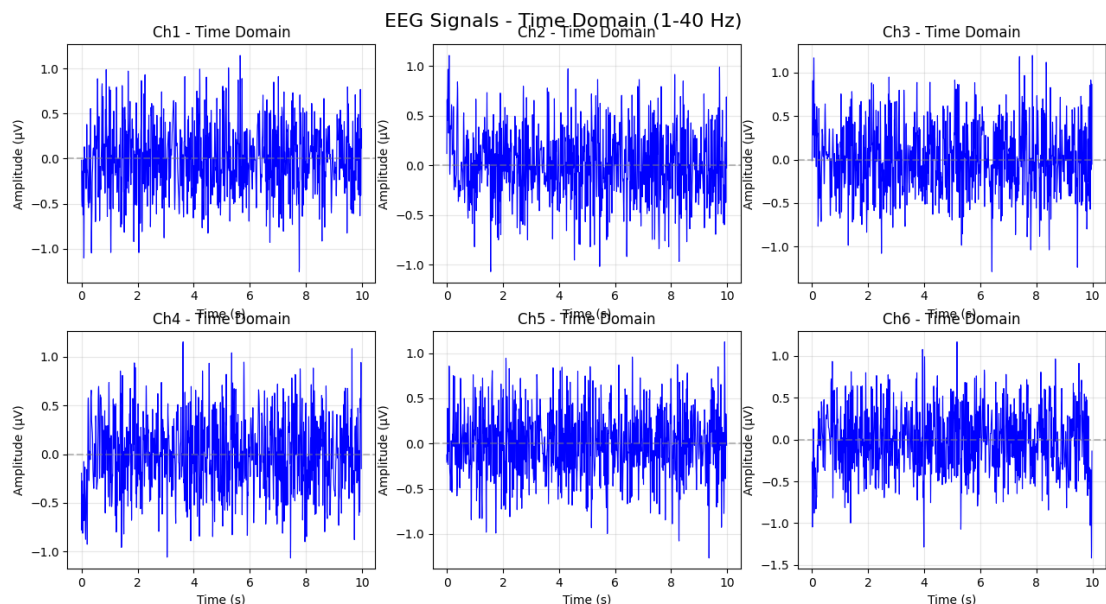


Fig 12. Short-circuit noise (monitor-only 1-40 Hz, 10 s, 6 channels).

The dataset size (trial level) is shown in Table 1 below:

Table 1. Dataset split for intent and emotion recognition tasks

Task	Train (trials)	Val (trials)	Test(trials)	Total
Intent	72	16	12	100
Emotion	42	9	9	60

4.2. Preprocessing and Feature-Level Signal Analysis

This section presents window sample construction, EEG signal distribution in time and frequency domains, channel-level statistics, and principal component analysis (PCA) based on concatenated features. All processing steps are performed after applying notch filtering, bandpass filtering, and z-score normalization. Detailed output logs and numerical results are provided in the **Appendix**.

4.2.1. Window Count and Class Distribution

Sliding windows with window size 500 samples (2.0 seconds) and hop size 250 samples (1.0 seconds) were applied. Feature extraction includes raw waveforms (time, shape = $n \times 6 \times 500$) and three-band power features ($\theta/\alpha/\beta$, shape = $n \times 6 \times 3$).

For window counts, intent tasks produced 648 training, 144 validation, and 108 test windows; emotion tasks produced 378, 81, and 81 windows respectively. Class proportions remain strictly balanced across all three splits, as shown in *Fig 13* and *Fig 14*. Each class in intent tasks (baseline, jaw clench, left gaze, right gaze) accounts for 25%, and each class in emotion tasks (negative, neutral, positive) accounts for 33.3%.

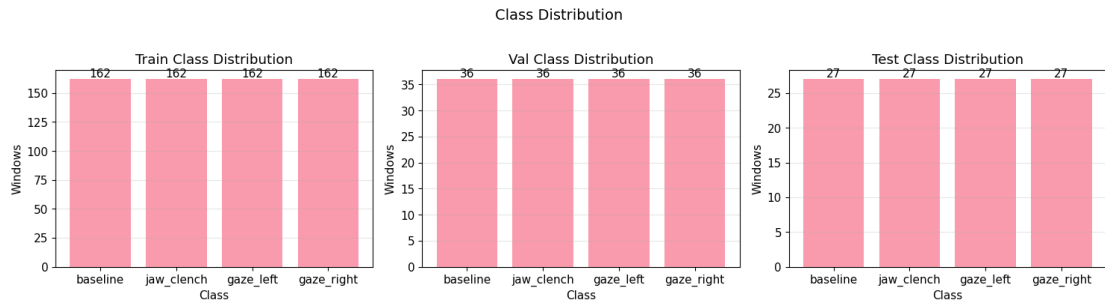


Fig 13. Class distribution after windowing (Intent)

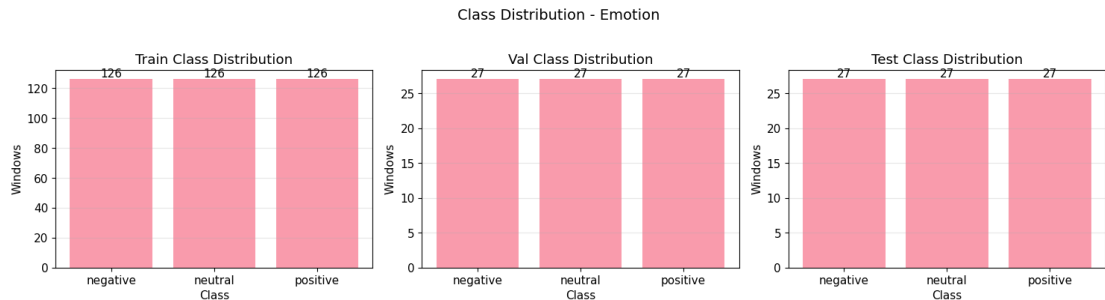


Fig 14. Class distribution after windowing (Emotion)

4.2.2. Time-Domain Signal Analysis

Training sets were grouped by class, computing mean $\pm$ standard error of the mean (SEM) for each channel to generate class-averaged waveforms. Fig 15 shows that in intent tasks, jaw clench class exhibits brief high-amplitude bursts in FT7/T3 and FT8/T4; gaze classes show low-frequency drift; baseline class shows lowest and most stable amplitude.

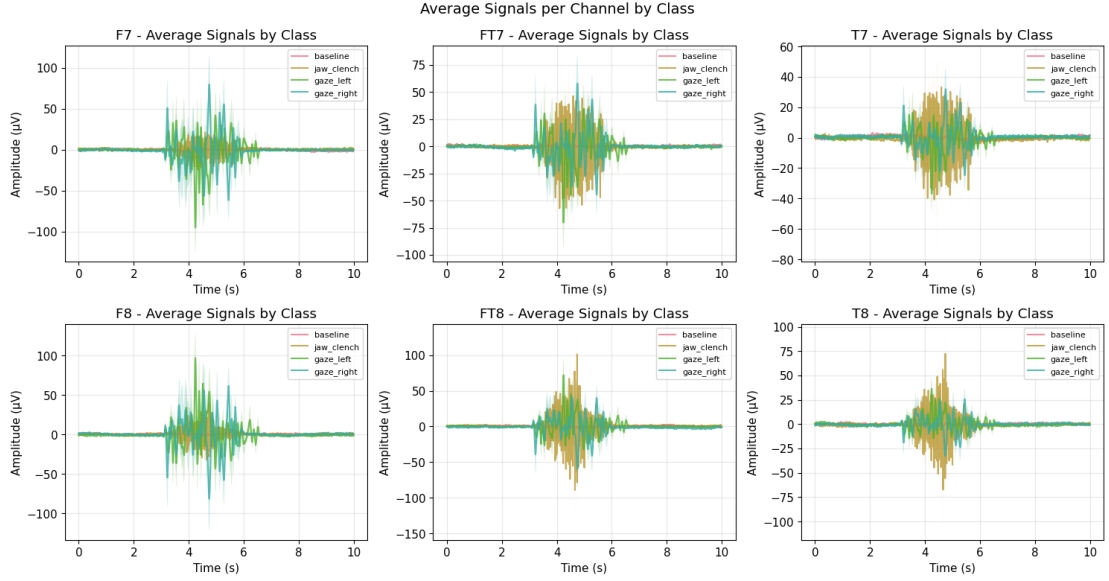


Fig 15. Average EEG signals by class (Intent, mean $\pm$ SEM)

Fig 16 presents average waveforms for emotion tasks, where all three classes have similar amplitude ranges on μ V scale, with class differences more reflected in rhythmic composition rather than instantaneous magnitude.

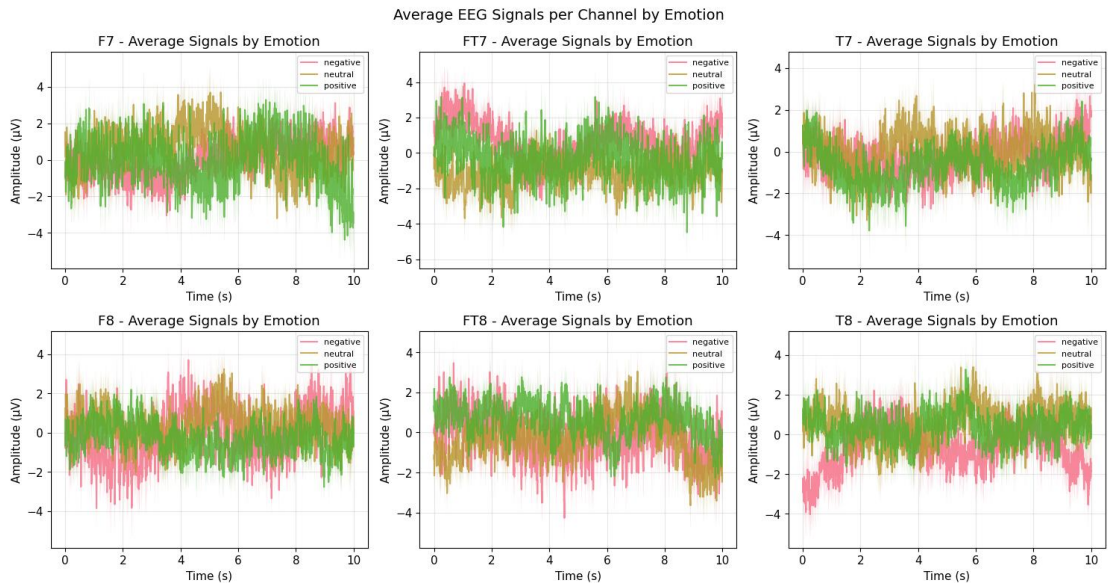


Fig 16. Average EEG signals by emotion (Emotion, mean $\pm$ SEM)

Note: T3 $\equiv$ T7 and T4 $\equiv$ T8 are used interchangeably in the figures; the plots adopt T7/T8.

4.2.3. Frequency-Band Power Features (Band-Power)

Band power in θ (4-8 Hz), α (8-13 Hz), and β (13-30 Hz) ranges was estimated per window using Welch's method, expressed as log power. Fig 17 and Fig 18 respectively summarize class distributions for intent and emotion tasks. In Fig 17, jaw clench class shows elevated β band power across channels, left/right gaze exhibit slightly higher θ band power, and baseline maintains lowest across all bands.

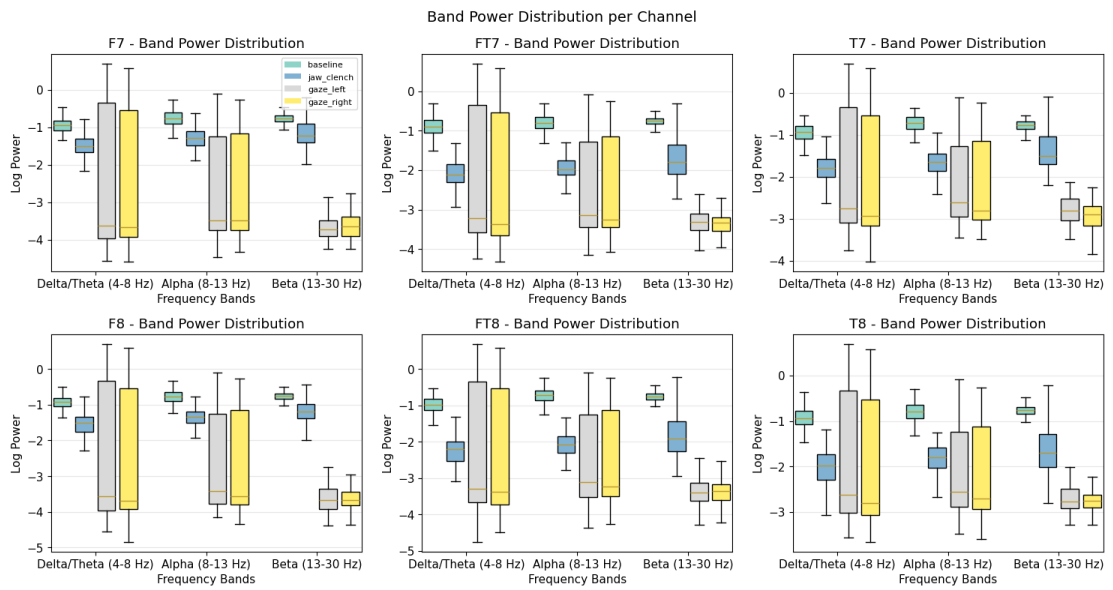


Fig 17. Band-power overview by class (Intent, $\theta/\alpha/\beta \times 6$ channels)

In Fig 18, α and β bands show consistent inter-class differences across channels. One-way ANOVA performed for each channel and band confirmed statistical significance for all combinations except θ band at T8 (traditional T4) in emotion tasks ($p = 0.1118$) (results provided in appendix).

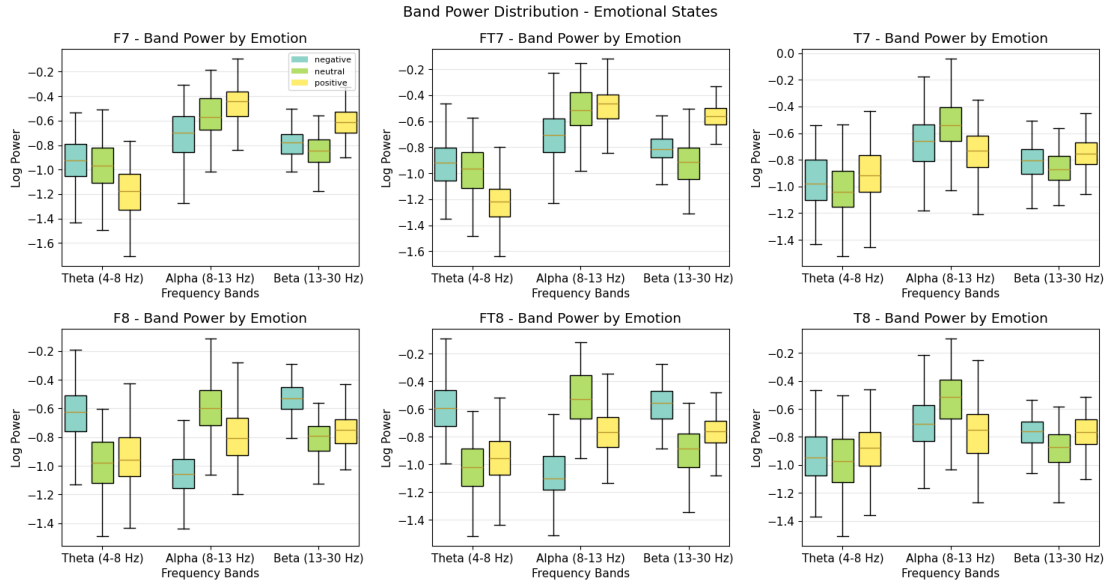


Fig 18. Band-power overview by emotion (Emotion, $\theta/\alpha/\beta \times 6$ channels)

4.2.4. Principal Component Analysis (PCA) of Feature Space

This section uses PCA to show global linear structure and t-SNE to examine local neighbourhoods; both are used only for visual analysis and do not participate in training or real-time inference.

As shown in Fig 19, in PCA projection of PC1 (3.5%) and PC2 (3.2%), baseline is compact near origin, jaw clench stretches along PC2, left and right gaze are broad and mutually overlapping with limited linear separability, indicating discriminative information is distributed across more principal components. In t-SNE plot, the two gaze regions still partially overlap, jaw clench forms a relatively dense band, and baseline is more scattered. This pattern is consistent with subsequently reported confusion matrices, explaining mutual confusion between left/right gaze and higher recall for jaw clench.

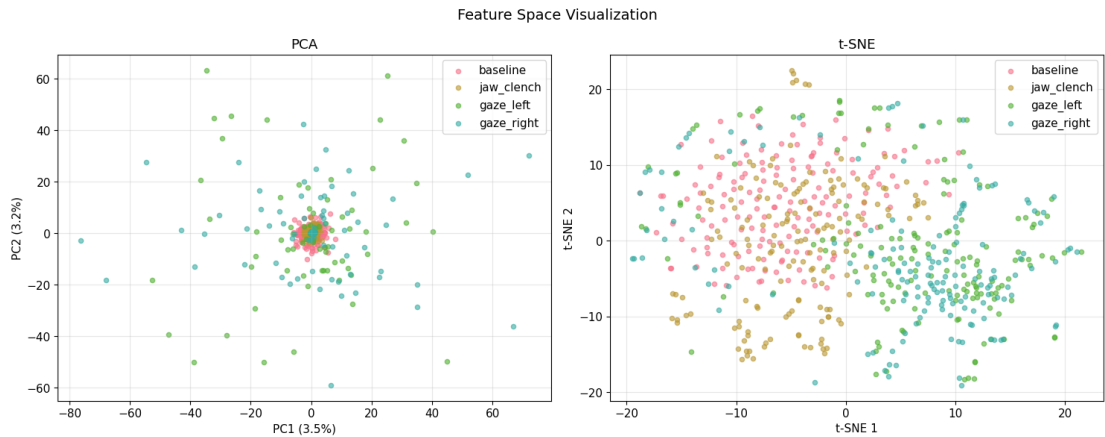


Fig 19. PCA projection and explained variance (Intent)

As shown in *Fig 20*, in PCA projection of PC1 (1.6%) and PC2 (1.5%), negative and positive substantially overlap while neutral is more scattered showing multiple sub-groups, reflecting weaker linear discriminative capability. In t-SNE plot, neutral shows as multiple "islands," negative and positive mix in centre, indicating intra-class heterogeneity and lower separability, consistent with larger variance across cross-validation folds (reported in Section 4.5.1).



Fig 20. PCA projection and explained variance (Emotion)

4.3. Model Ablation and Architecture Selection

4.3.1. Intent Task Architecture Selection

For the intent task, *Fig 21* compares the validation accuracy of four alternative architectures. The multimodal fusion model (temporal + frequency) achieved the highest accuracy of 76.39%, outperforming the temporal-only (72.92%), frequency-only (68.06%), and image-only (30.56%) networks. Owing to this superior performance, the multimodal network was selected as the final model structure for subsequent training and evaluation.

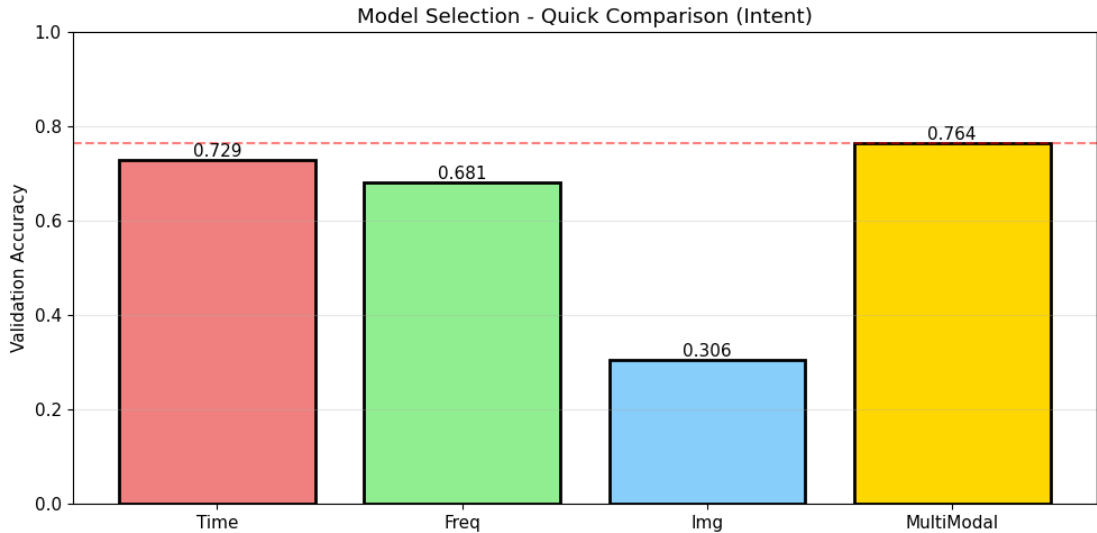


Fig 21. Validation Accuracy of Different Model Structures (Intent)

4.3.2. Emotion Task Architecture Selection

For the emotion task, Fig 22 shows that the frequency-only architecture attained the best validation accuracy at 76.54%, followed by the multimodal network (67.90%), the image-only model (59.26%), and the temporal-only model (43.21%). Given its higher accuracy and lower complexity, the frequency-only architecture was chosen for training and deployment.

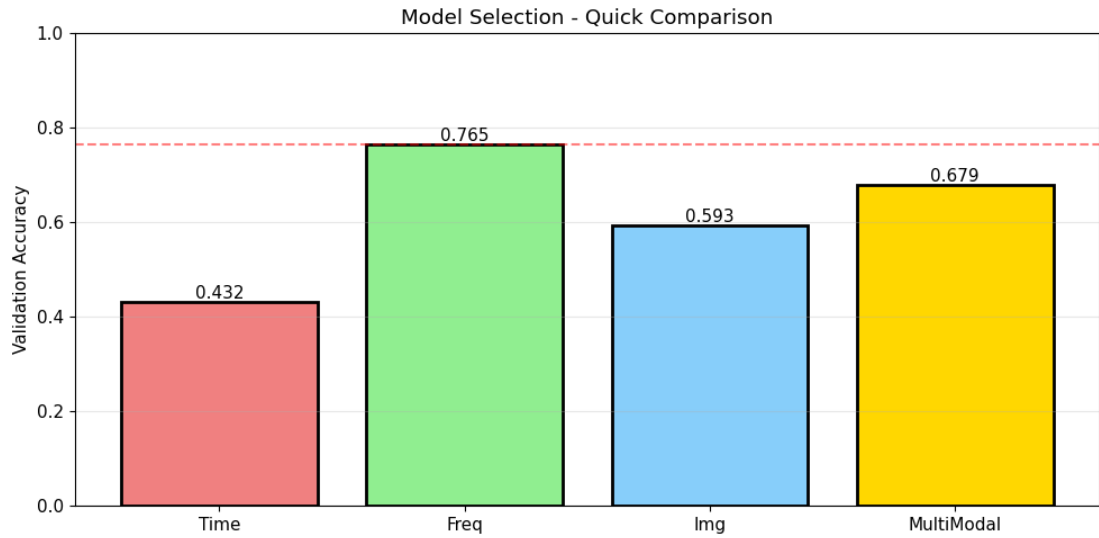


Fig 22. Validation Accuracy of Different Model Structures (Emotion)

4.4. Model Training and Evaluation

This section presents training processes and test performance of selected models, including training curves, confusion matrices, and classification reports.

4.4.1. Training Process

Fig 23 shows loss and accuracy curves for the intent model over 50 epochs. Best validation accuracy (75.69%) was reached at epoch 17, at which point the model was saved for evaluation.

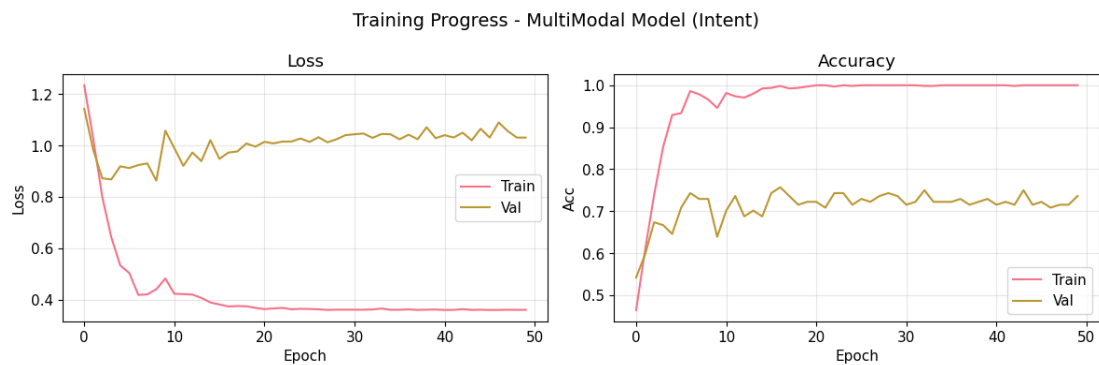


Fig 23. Training Curves (Loss and Accuracy) (Intent)

Fig 24 shows the training process for the emotion model. The frequency-only model reached peak validation accuracy of 74.07% at epoch 45, with the corresponding model saved for testing.



Fig 24. Training Curves (Loss and Accuracy) (Emotion)

4.4.2. Test Set Performance

As shown in Table 2 and Fig 25, baseline class was perfectly recognized. Jaw clench achieved 0.85 recall, while left and right gaze frequently confused with each other, leading to lower precision and F1 scores.

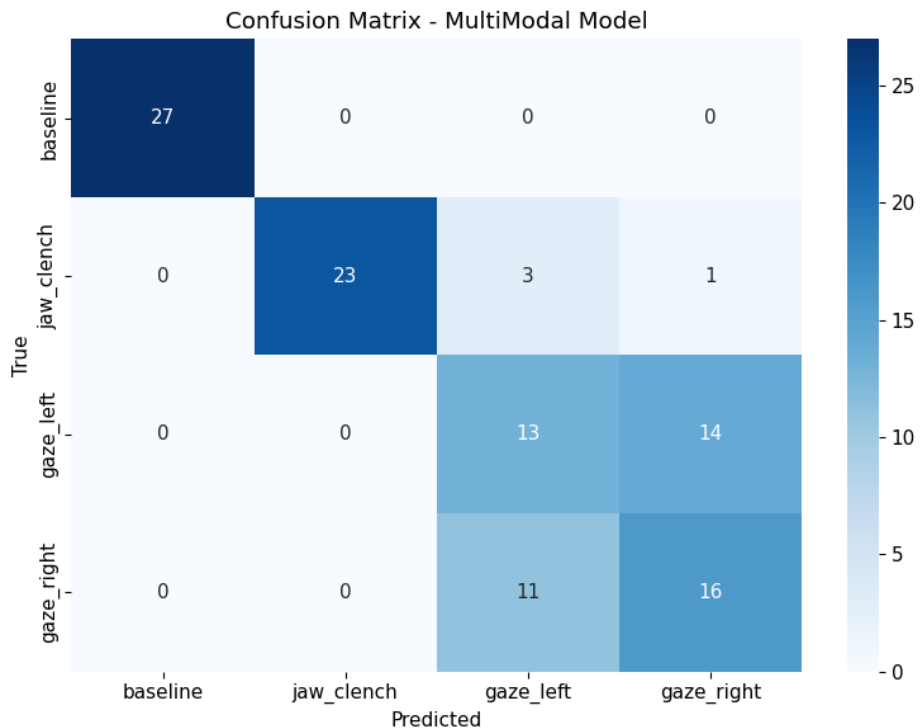


Fig 25. Confusion Matrix - Intent Task (Test Set)

Table 2. Classification report for intent recognition (test set, N=108)

Class	Precision	Recall	F1-score	Support
Baseline	1	1	1	27
Jaw Clench	1	0.8519	0.92	27
Gaze Left	0.4815	0.4815	0.4815	27
Gaze Right	0.5161	0.5926	0.5517	27
Accuracy	/	/	0.7315	108
Macro Avg	0.7494	0.7315	0.7383	108
Weighted Avg	0.7494	0.7315	0.7383	108

Table 3 and *Fig 26* show the emotion classification report. Negative valence reached the highest recall at 0.89, while neutral was the most difficult class with a recall of 0.44. Positive class reached a recall of 0.70 but lower precision.

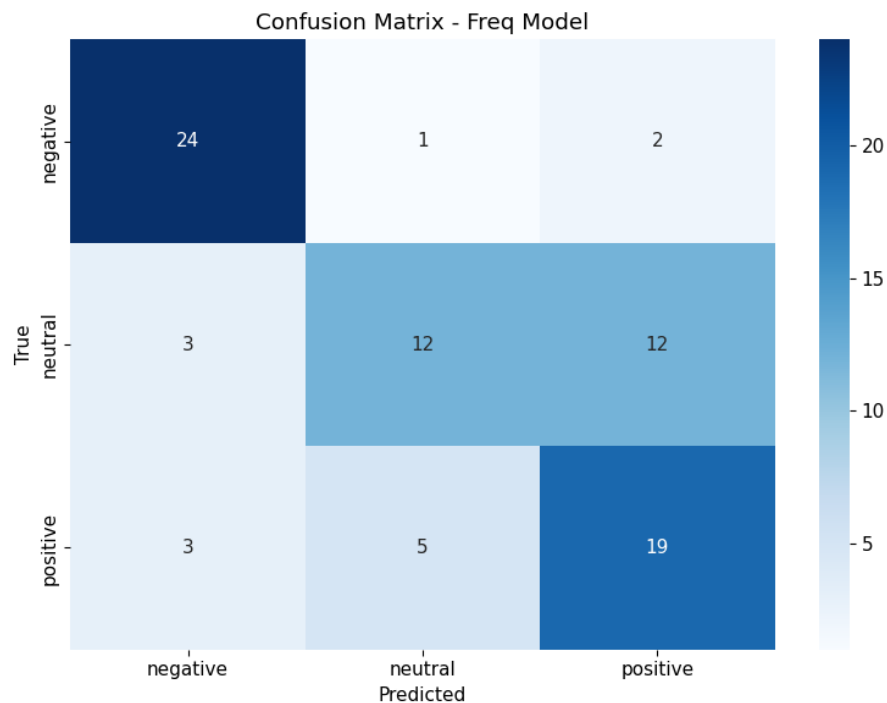


Fig 26. Confusion Matrix - Emotion Task (Test Set)

Table 3. Classification report for emotion recognition (test set, N=81)

Class	Precision	Recall	F1-score	Support
Negative	0.8	0.8889	0.8421	27
Neutral	0.6667	0.4444	0.5333	27
Positive	0.5758	0.7037	0.6333	27
Accuracy	/	/	0.679	81
Macro Avg	0.6808	0.679	0.6696	81
Weighted Avg	0.6808	0.679	0.6696	81

4.4.3. Overall Evaluation Metrics

Table 4 summarises the test set metrics for both tasks. Intent recognition achieved higher performance, with substantial agreement indicated by Cohen's Kappa (0.64) and a practical ITR of 44.1 bits/min. Emotion recognition remained more challenging, reaching moderate agreement (Kappa 0.52) and an ITR of 21.5 bits/min.

Table 4. Summary of quantitative results for intent and emotion recognition

Task	Accuracy	Macro-F1	Cohen's Kappa	Balanced Accuracy	ITR (bits/trial)	ITR (bits/min)
Intent	0.7315	0.7383	0.642	0.7315	0.735	44.1
Emotion	0.679	0.6696	0.5185	0.679	0.359	21.5

4.4.4. Feature Importance Analysis

To further understand model decision mechanisms across different modalities, this section applies gradient-based backpropagation and occlusion analysis to extract sensitivity distributions of trained models relative to input features.

In time-domain models, Fig 27 illustrates temporal distribution of feature importance for six channels in intent tasks. Channels FT8 and FT7 show highest sensitivity in the 0.5-1.5 second range, further confirming the contribution of transient high-amplitude activity related to jaw clench class to model predictions.

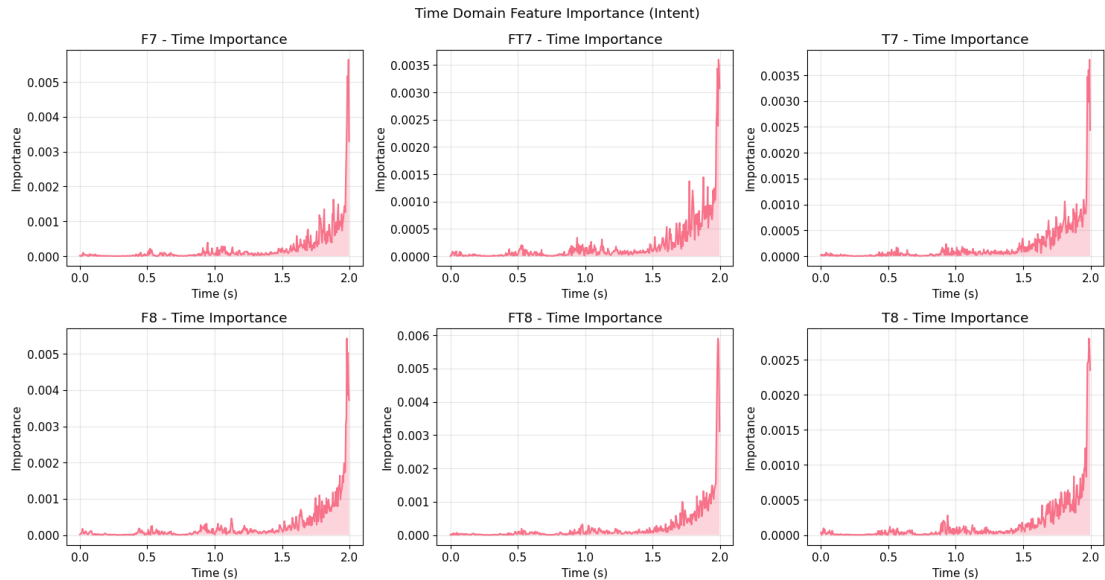


Fig 27. Time Domain Feature Importance (Intent)

In the frequency-domain models, Fig 28 and Fig 29 respectively show importance weights of θ , α , and β bands across six channels for intent and emotion

tasks. For intent tasks, channel FT8 contributes most in θ and β bands. In emotion tasks, β bands of F8 and FT8 are most prominent, indicating that emotional rhythms in right temporal regions are particularly relevant to classification.

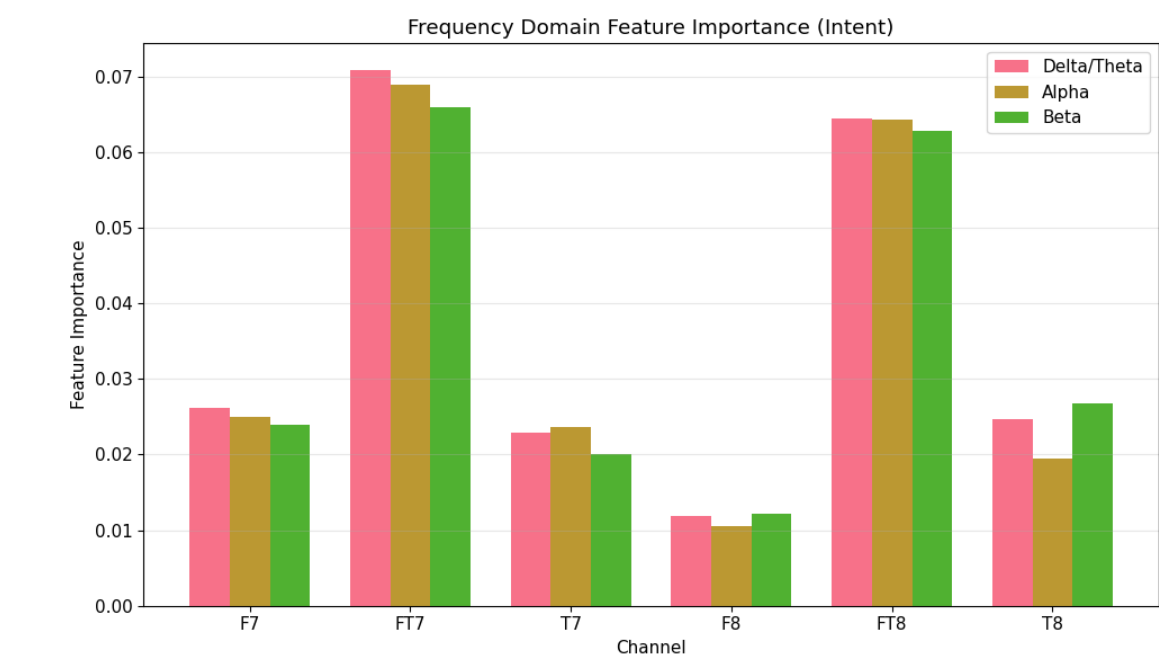


Fig 28. Frequency Band Importance per Channel (Intent)

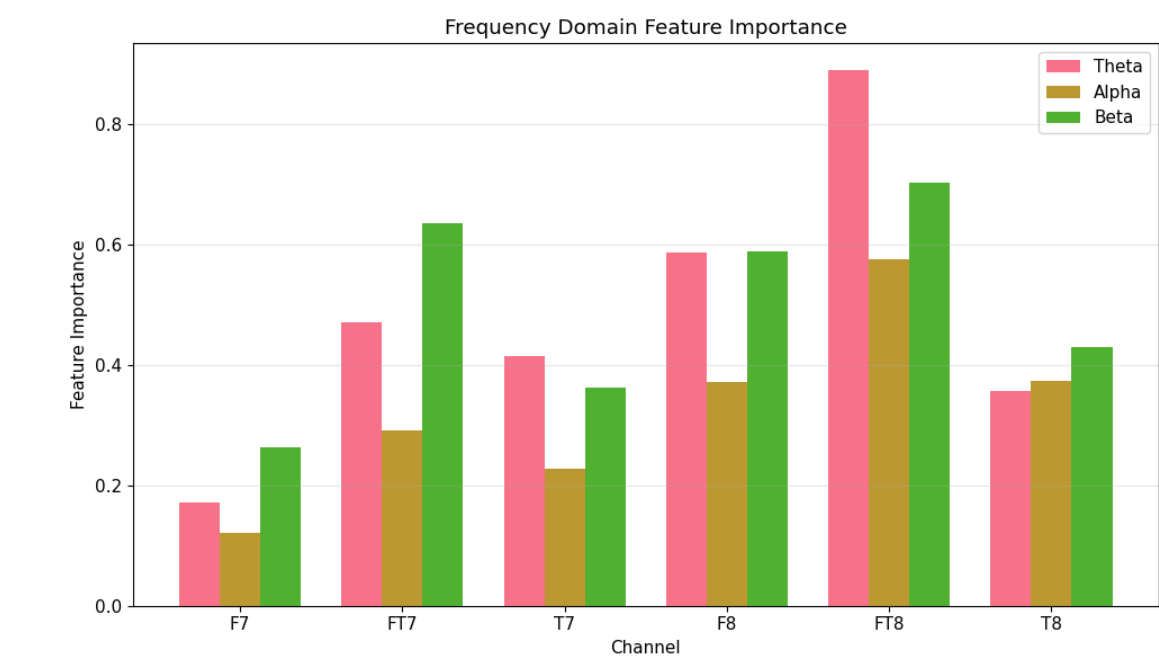


Fig 29. Frequency Band Importance per Channel (Emotion)

4.5. Cross-Validation and Real-Time Inference Performance

4.5.1. Five-Fold Cross-Validation

Five-fold cross-validation was performed on intent and emotion tasks to evaluate

model robustness. For intent tasks, accuracy ranged from 71.67% to 80.56% with mean 76.33% and standard deviation 3.42%. For emotion tasks, accuracy varied between 68.52% and 88.89% with mean 75.37% and standard deviation 7.09%. These results indicate both models maintain stable performance even under limited data conditions.

4.5.2. Real-Time Inference Performance

Real-time evaluation uses 2.0-second windows and 1-second update intervals; reported inference times exclude I/O and buffering overhead, while end-to-end latency includes complete delay from acquisition to decision.

As shown in *Fig 30*, intent model achieves average inference time 3.40 ± 0.55 ms, throughput approximately 294 Hz, corresponding to a real-time factor (RTF) of 588.6.

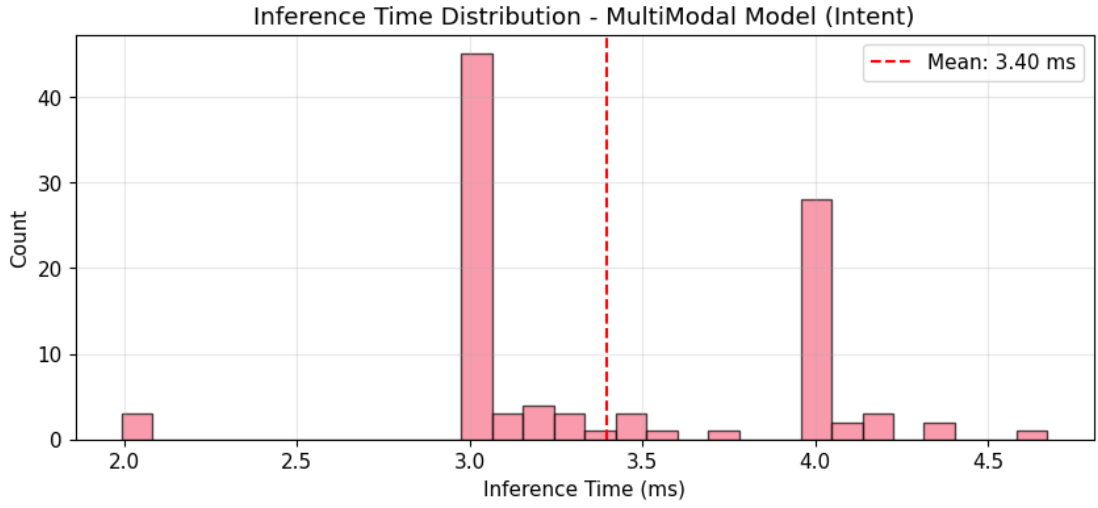


Fig 30. Inference Time Distribution and Real-Time Performance (Intent)

And as shown in *Fig 31*, emotion model achieves average inference time 0.16 ± 0.39 ms, throughput approximately 6125 Hz, RTF 12250.6×.

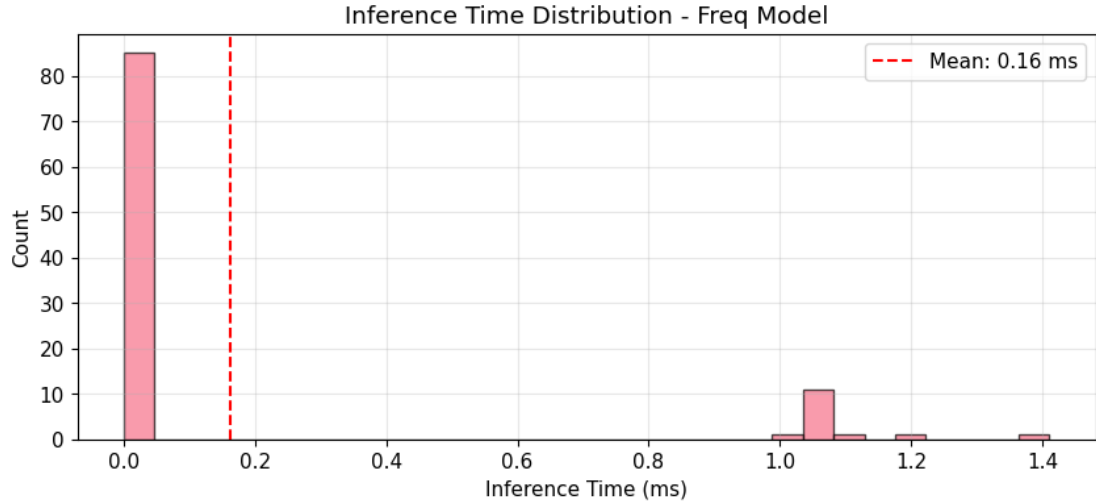


Fig 31. Inference Time Distribution and Real-Time Performance (Emotion)

Both models significantly exceed real-time requirements, demonstrating suitability for low-latency BCI applications.

5. Discussion

5.1. Overview and Alignment with Objectives

This study proposes a compact glasses-based bioelectric acquisition and inference system using six dry electrode channels at 250 Hz sampling rate to acquire signals and achieve real-time decoding on embedded hardware. The system processes two core tasks: discrete intent recognition (baseline, jaw clench, left gaze, right gaze) and three-class emotion recognition (negative, neutral, positive).

Under configurations of 2.0-second analysis windows and 1-second update intervals, the system achieves practical classification accuracy with millisecond-level inference latency. Intent recognition achieved balanced accuracy of 73.15% with information transfer rate of 44.1 bits/min; emotion recognition achieved 67.90% accuracy but with more significant inter-class differences. Overall, these results validate system feasibility and achieve this study's objectives in transparency, end-to-end design, and real-time behaviour quantification.

5.2. Interpreting Results by Modality and Class

Performance differences are highly correlated with modality dominance.

Intent recognition: Jaw clench belongs to EMG-dominated actions, manifesting

as high-amplitude, short-latency bursts that can be stably captured through time-domain features while introducing broadband energy in β bands as a byproduct. Baseline presents as low-variance signals after Z-score normalization, thus achieving high specificity. Lateral gaze belongs to EOG-dominated actions, mainly presenting low-frequency eye deflections with certain left-right symmetry. Limited spatial sampling and symmetry result in residual confusion between left and right gaze.

Emotion recognition: Mainly relies on EEG rhythmic signals. However, due to short analysis windows and electrodes covering only six frontotemporal channels, spatial resolution and temporal integration capabilities are limited, thus weakening discrimination of α/β bands and producing large overlap around neutral class.

Confusion matrices reflect consistency with physiological mechanisms: in intent recognition, errors mainly concentrate between left and right gaze, while jaw clench and baseline are more robust due to their significant EMG features and stable resting states. In emotion recognition, errors mainly concentrate on neutral class, as neutral naturally lies between negative and positive and spans different arousal levels [49]. Model confidence decline appears as expected in this case, suggesting future introduction of confidence-aware logic to optimize downstream decisions.

5.3. Feature-Architecture Rationale

Ablation experiment results validate the rationality of task-specific architectures.

Intent tasks: Best results come from late fusion of temporal and frequency features. Time-domain features can capture EMG bursts and EOG deflections, while frequency-domain features summarize relatively stable slow rhythms; their combination provides complementary information under controlled model capacity. Spectrogram branch failed to improve generalization and added computational overhead and latency jitter in real-time systems, thus was disabled.

Emotion tasks: Frequency-only features performed best. This is consistent with EEG dominance in emotion recognition, where rhythmic components (particularly α/β balance) are more discriminative than transient morphological patterns. This design both reduces overfitting risk in small datasets and fits embedded hardware resource constraints.

Time window selection: 2.0-second context with 1-second update intervals (50% overlap) achieves balance between responsiveness and accuracy. Shorter steps could reduce perceptual latency but increase variance due to insufficient averaging; longer context might improve stability but reduce interaction sensitivity and delay feedback. This study's parameter choice ensures stable decision refresh

rates without overloading the computational budget.

5.4. Real-Time Feasibility and Systems Engineering

Measured inference times show that intent recognition model latency is in the low millisecond range, while the lightweight emotion recognition model latency is even sub-millisecond, leaving ample margin for embedded processors. However, overall perceptual latency depends not only on inference but also on buffer accumulation, filter group delay, inter-process communication, visualization refresh, and actuator latency.

To minimize these overheads, the system adopts a streaming-oriented design. Key elements include fixed-size ring buffers, efficient IIR/FIR filters, and single-copy feature paths. Disabling the spectrogram branch avoids GPU-related computational uncertainty, thus stabilizing frame times. Computing EMG/EOG-dominated temporal features and EEG-oriented frequency features synchronously within the same 2.0-second context avoids cross-modal latency offsets.

Therefore, the entire processing pipeline exhibits low-jitter, predictable characteristics suitable for interactive tasks. From a systems engineering perspective, co-locating acquisition and inference on the same System-on-Chip avoids USB/Ethernet transmission overhead and reduces instability from clock domain mismatches. Further, employing deterministic scheduling (like timer-driven callbacks) and lock-free queue mechanisms effectively improves system real-time performance.

5.5. Limitations

5.5.1. Technical Limitations

Cross-modal signal leakage: EMG and EOG activity may contaminate EEG frequency ranges, while current pipeline employs no explicit separation strategies beyond filtering and normalization. Therefore, α/β power used for emotion recognition may carry non-neural components, bringing misattribution risks. This problem is particularly prominent during jaw clenching or large saccades, as broadband EMG/EOG energy extends to high-frequency ranges.

Impedance drift over time: Electrode-skin impedance is not monitored or compensated across sessions. Movement, sweating, or electrode position micro-adjustments may cause impedance changes, introducing amplitude bias and non-stationarity, thus weakening stability and long-term performance of six-channel frontotemporal electrode layout.

Temporal resolution limitations: Uniform use of 2.0-second windows and 1.0-second update intervals helps offline and online result consistency but sacrifices temporal acuity. Rapid emotional fluctuations or tightly time-locked phenomena (like P300, error-related potentials) may be averaged or masked, limiting system applicability in fast event-related BCI paradigms.

Emotional label subjectivity: Intent labels are clear and unambiguous, while emotion labels are inherently subjective. Due to lack of multi-rater annotation or physiological cross-validation (like EDA, PPG, pupil response), valence/arousal annotations may contain noise. This uncertainty limits achievable model accuracy and increases variance, especially under neutral conditions.

Hardware and interference sensitivity: Acquisition relies on wired connections, with electromagnetic sensitivity and mechanical noise characteristics not systematically evaluated. Environmental interference and movement may produce differential impacts across sessions, thus exacerbating signal non-stationarity.

5.5.2. Methodological Limitations

Lack of power analysis: No prospective power calculation was performed to determine sample size needed to detect target effects. Therefore, result differences may be limited by small sample variance rather than statistical adequacy, and confidence intervals and dispersion metrics should be interpreted cautiously.

Single-participant data overfitting risk: Although cross-validation was used, all data came from a single participant, and models may over-fit individual characteristics, exaggerating apparent generalization capability. Lack of multi-participant validation or cross-session evaluation leaves cross-individual performance unknown.

Lack of online adaptation and drift handling: Real-time systems do not include drift detection, recalibration triggers, or adaptive normalization mechanisms. During long-term use, cross-day experiments, or device re-wearing, performance may decline due to lack of dynamic adjustment.

5.5.3. Validation Limitations

Lack of closed-loop validation: This study's evaluation focused on open-loop prediction accuracy without conducting user-in-the-loop experiments. Therefore, actual interaction metrics like task completion time, error recovery, subjective load, and usability have not been quantified, leaving user experience and actual controllability uncertain.

Limited ecological validity: Experiments were conducted in controlled environments without testing effects of daily activities and interference (like walking, head movement, speech, facial expressions, environmental light/noise), leaving system applicability in natural scenarios unverified.

Wired acquisition only: Current prototype achieves real-time operation through cables, which satisfies proof-of-concept but lacks sufficient wearability. Wireless applications would require addressing packet loss, jitter, and RF coexistence issues, which are not covered in this study.

5.6. Potential Application Scenarios

5.6.1. AR/VR Interaction

This system can naturally map to hands-free interaction scenarios in head-mounted displays. A practical design is a hybrid selector: camera-based eye tracking provides rapid positioning in 2D space, while biosignal intent (like EMG-dominated jaw clench or EEG-based confirmation) serves as neural "click" or mode switching to alleviate false trigger issues. Lateral gaze classes can still serve as redundancy support and provide fallback mechanisms when eye tracking fails.

To be compatible with 90-120 Hz UI refresh rates, system stacks need to meet strict end-to-end latency budgets by minimizing buffer lengths, co-locating inference, and prioritizing constant-time feature extraction.

5.6.2. Affective Media Personalization

Emotion estimation can be used for on-device adjustment of music and short video recommendations. Contextual bandit algorithms combine predicted valence/arousal with explicit feedback to limit exploration risks while preserving features locally for privacy protection. Stronger and validated stimuli, along with brief on-site self-reports, can provide support for user calibration; meanwhile, auxiliary physiological signals (like EDA, PPG, pupil dynamics) can provide weak supervision to stabilize labels without requiring extensive manual annotation.

5.6.3. Assistive Smart Homes

Robust intent recognition classes can provide discrete device control for elderly or motor-impaired users. Mapping to scene presets through Matter or Zigbee protocols enables low-latency control across vendors. Safety aspects require rule-based protection measures, such as multi-hop confirmation for critical actions, sustained intent thresholds, and immediate revocation functions. Meanwhile, multimodal physiological signal fallback mechanisms (voice, physical buttons) remain crucial for improving system resilience.

6. Conclusion

This study designed and validated a glasses-based brain-computer interface system capable of acquiring six-channel dry electrode bioelectric signals (EEG/EOG/EMG) at 250 Hz sampling rate and achieving real-time decoding on embedded hardware. With 2.0-s windows and 1.0-s updates, the system achieved 73.15% test accuracy for intent recognition (44.1 bits/min) and 67.90% for emotion recognition (21.5 bits/min), with millisecond-level inference latency on embedded hardware.

Ablation studies further validated the rationality of task-specific architectures: intent recognition optimally used late fusion of temporal and frequency features, capable of simultaneously capturing EMG-dominated jaw clench transients and EOG-driven eye movement deflections, enhanced by rhythmic summaries; emotion recognition showed optimal performance with frequency-only feature models, reflecting this task's primary reliance on EEG rhythms, especially α/β band composition.

Results also revealed why intent recognition outperforms emotion recognition. EMG and EOG provide higher signal-to-noise ratio features, making jaw clench and lateral gaze more distinguishable; while emotion recognition relies on EEG rhythms that show large heterogeneity under short time windows and limited spatial coverage, particularly with obvious overlap around neutral class. Real-time evaluation showed that the system retained sufficient computational margin. However, end-to-end user interaction latency depends on the entire closed-loop budget—including buffering, filtering, inter-process communication, and UI updates—which imposes strict requirements on streaming design to meet perceptual thresholds for responsiveness.

The study also has certain constraints. Dataset size is small and comes from only a single participant, limiting statistical power and external validity. Electrode number and positions are constrained by glasses frame, and contact stability affects signal quality. Acquisition is wired and sensitive to environmental interference. Emotion elicitation stimuli may be insufficient to cover stronger emotional states, and vertical eye movement control is not yet supported. Cross-modal leakage (EMG/EOG into EEG bands) may affect rhythmic interpretation, especially prominent without artifact-aware training.

Future application prospects are noteworthy. In AR/VR scenarios, the system can serve as hybrid interaction: camera-based eye tracking for 2D aiming with biosignal intent as "neural confirmation" channels, meeting strict low-latency

interaction requirements [50]. In emotional media, on-device emotion estimation can personalize music or video recommendations through confidence exposure and fallback mechanisms to ensure experience. In smart homes, robust intent recognition can be used for discrete device control, combined with safety protection rules and multimodal redundancy channels to improve system reliability [51].

Next steps focus on expanding multi-participant, multi-session datasets, introducing self-supervised pretraining and lightweight temporal modelling networks to improve generalization. Meanwhile, wireless and mechanical design improvements are needed to achieve truly daily wearable forms, and stronger stimuli and auxiliary physiological signals are needed to improve emotion label quality. These developments will drive prototype systems toward practical deployment, laying technical foundations for future human-computer interaction based on emotion perception and intent recognition.

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Appendix

All code and notebook outputs are available at:

https://github.com/GanshinW/Glasses_Based_BCI

Intent_pipeline.ipynb output:

Dataset loaded

Shapes | X_train(72, 6, 2500) X_val(16, 6, 2500) X_test(12, 6, 2500)

Meta | fs=250Hz ch=6 classes=['baseline', 'jaw_clench', 'gaze_left', 'gaze_right']

Preprocess: notch/bandpass/z-score done

Windowing: size=500 hop=250 -> Train648, Val144, Test108

Feature extraction: time+bandpower done

Features | time_train(648, 6, 500) freq_train(648, 6, 3)

Class balance (%):

Train baseline:25.0%, jaw_clench:25.0%, gaze_left:25.0%, gaze_right:25.0%

Val baseline:25.0%, jaw_clench:25.0%, gaze_left:25.0%, gaze_right:25.0%

Test baseline:25.0%, jaw_clench:25.0%, gaze_left:25.0%, gaze_right:25.0%

Channel stats (Train):

Channel	Mean±SD (μV)	Min	Max	
SNR(dB)				
F7	-0.44±42.23	-717.03	646.05	0.00
FT7	-0.38±35.10	-855.12	711.31	0.00
T7	-0.04±21.40	-656.97	468.55	0.00
F8	0.55±42.77	-644.56	759.36	0.00
FT8	0.05±36.19	-821.74	779.25	0.00
T8	-0.24±22.44	-628.78	651.17	0.00

Band-power ANOVA p-values (p<0.05 indicates significance)

Channel	Delta/Theta	Alpha	Beta
---------	-------------	-------	------

F7	0.0000	0.0000	0.0000
FT7	0.0000	0.0000	0.0000
T7	0.0000	0.0000	0.0000
F8	0.0000	0.0000	0.0000
FT8	0.0000	0.0000	0.0000
T8	0.0000	0.0000	0.0000

PCA: 10 comps=26.7%, 50 comps=67.6%, 90% at 1 comps

=== MODEL SELECTION ===

Testing Time-only...

Time-only val acc: 0.7292

Testing Freq-only...

Freq-only val acc: 0.6806

Testing Img-only...

Img-only val acc: 0.3056

Testing MultiModal...

MultiModal val acc: 0.7639

*** Best model: MultiModal with val acc: 0.7639 ***

=== FULL TRAINING WITH MULTIMODAL MODEL ===

Device: cuda

Params: total=11,270,724 trainable=11,270,724

Training ...

Epoch 10/50 | Train 0.4827/0.9460 | Val 1.0577/0.6389

Epoch 20/50 | Train 0.3685/0.9969 | Val 0.9952/0.7222

Epoch 30/50 | Train 0.3615/1.0000 | Val 1.0401/0.7361

Epoch 40/50 | Train 0.3619/1.0000 | Val 1.0283/0.7292

Epoch 50/50 | Train 0.3608/1.0000 | Val 1.0305/0.7361

Train done

Best val acc: 0.7569 @ epoch 17

Model saved: ../best_intent_multimodal_model.pth

Test | loss=1.0205 acc=0.7315

Classification report:

	precision	recall	f1-score	support
baseline	1.0000	1.0000	1.0000	27
jaw_clench	1.0000	0.8519	0.9200	27
gaze_left	0.4815	0.4815	0.4815	27
gaze_right	0.5161	0.5926	0.5517	27
accuracy			0.7315	108
macro avg	0.7494	0.7315	0.7383	108
weighted avg	0.7494	0.7315	0.7383	108

Extra metrics | acc=0.7315 macro-F1=0.7383 kappa=0.6420
bAcc=0.7315 ITR=0.735 bits/trial, 44.1 bits/min (T=1.0s)

RT perf:

infer=3.40±0.55 ms (min=1.99, max=4.67) | rate≈294.3 Hz | win=2.0s |
RTF=588.6x

=== CROSS-VALIDATION ===

CV (5-fold) with MultiModal model...

Fold 1: 0.7944

Fold 2: 0.7167

Fold 3: 0.8056

Fold 4: 0.7667

Fold 5: 0.7333

CV (5-fold): mean=0.7633±0.0342 | best=0.8056 | worst=0.7167

=====

=== FINAL SUMMARY - INTENT DETECTION ===

=====

Dataset | windows=900 trials=100 classes=4 fs=250Hz win=2.0s
ch=6

Model Selection (Step 8):

Time val_acc=72.92%

Freq val_acc=68.06%

Img val_acc=30.56%

*** MultiModal val_acc=76.39%

Final Model Performance (MultiModal):

Train/Val	best_val=75.69% @ epoch 17
Test	acc=73.15%
Cross-Val	mean=76.33%±3.42% (5-fold)

Overall Metrics:

Accuracy	0.7315
Macro-F1	0.7383
Cohen's Kappa	0.6420
Balanced Acc	0.7315
ITR	0.735 bits/trial, 44.1 bits/min

Real-time Performance:

Inference	3.40±0.55 ms
Throughput	294.3 Hz
RTF	588.6x real-time

Emotion_pipeline.ipynb output:

Dataset loaded

Shapes | X_train(42, 6, 2500) X_val(9, 6, 2500) X_test(9, 6, 2500)

Meta | fs=250Hz ch=6 classes=['negative', 'neutral', 'positive']

Preprocess: notch/bandpass/z-score done

Windowing: size=500 hop=250 -> Train378, Val81, Test81

Feature extraction: time+bandpower done

Features | time_train(378, 6, 500) freq_train(378, 6, 3)

Class balance (%):

Train negative:33.3%, neutral:33.3%, positive:33.3%

Val negative:33.3%, neutral:33.3%, positive:33.3%

Test negative:33.3%, neutral:33.3%, positive:33.3%

Channel stats (Train):

Channel	Mean±SD (µV)	Min	Max	
SNR(dB)				
F7	0.21±4.60	-19.22	20.51	0.01
FT7	-0.24±4.58	-19.92	19.81	0.01
T7	-0.27±4.22	-15.77	21.08	0.02
F8	0.02±4.40	-20.45	18.65	0.00
FT8	0.04±4.44	-20.16	19.13	0.00
T8	0.09±4.40	-17.06	16.84	0.00

Band-power ANOVA p-values (p<0.05 indicates significance)

Channel	Theta	Alpha	Beta
F7	0.0000	0.0000	0.0000
FT7	0.0000	0.0000	0.0000
T7	0.0002	0.0000	0.0000
F8	0.0000	0.0000	0.0000
FT8	0.0000	0.0000	0.0000
T8	0.1118	0.0000	0.0000

PCA: 10 comps=12.8%, 50 comps=41.5%, 90% at 1 comps

Testing Time-only model...

Time-only val acc: 0.4321
Testing Freq-only model...
Freq-only val acc: 0.7654
Testing Img-only model...
Img-only val acc: 0.5926
Testing MultiModal...
MultiModal val acc: 0.6790

*** Best model: Freq with val acc: 0.7654 ***

Device: cuda
Params: total=2,531 trainable=2,531
Training ...
Epoch 10/50 | Train 1.0885/0.6614 | Val 1.0864/0.6173
Epoch 20/50 | Train 1.0652/0.6746 | Val 1.0669/0.6790
Epoch 30/50 | Train 1.0097/0.6931 | Val 1.0028/0.7037
Epoch 40/50 | Train 0.9146/0.7143 | Val 0.9011/0.7037
Epoch 50/50 | Train 0.8258/0.7275 | Val 0.8064/0.7407

Train done
Best val acc: 0.7407 @ epoch 45
Model saved: ../best_emotion_freq_model.pth

Test | loss=0.9032 acc=0.6790

Classification report:

	precision	recall	f1-score	support
negative	0.8000	0.8889	0.8421	27
neutral	0.6667	0.4444	0.5333	27
positive	0.5758	0.7037	0.6333	27
accuracy			0.6790	81
macro avg	0.6808	0.6790	0.6696	81
weighted avg	0.6808	0.6790	0.6696	81

Extra metrics | acc=0.6790 macro-F1=0.6696 kappa=0.5185
bAcc=0.6790 ITR=0.359 bits/trial, 21.5 bits/min (T=1.0s)

RT perf:

infer=0.16±0.39 ms (min=0.00, max=1.41) | rate≈6125.3 Hz | win=2.0s |

RTF=12250.6x

=== CROSS-VALIDATION ===

CV (5-fold) with Freq model...

Fold 1: 0.6852

Fold 2: 0.7130

Fold 3: 0.7315

Fold 4: 0.7500

Fold 5: 0.8889

CV (5-fold): mean=0.7537±0.0709 | best=0.8889 | worst=0.6852

=====

=== FINAL SUMMARY - EMOTION RECOGNITION ===

=====

Dataset | windows=540 trials=60 classes=3 fs=250Hz win=2.0s
ch=6

Model Selection Results (Step 8):

Time	val_acc=43.21%
*** Freq	val_acc=76.54%
Img	val_acc=59.26%
MultiModal	val_acc=67.90%

Final Model Performance (Freq):

Training | best_val=74.07% @ epoch 45

Test | acc=67.90%

Cross-Val | mean=75.37%±7.09% (5-fold)

Per-Class Performance:

Overall Metrics:

Accuracy | 0.6790

Macro-F1 | 0.6696

Cohen's Kappa | 0.5185

Balanced Acc | 0.6790
ITR | 0.359 bits/trial, 21.5 bits/min

Real-time Performance:

Inference | 0.16 ± 0.39 ms
Throughput | 6125.3 Hz
RTF | 12250.6x real-time